HYBRID ADAPTIVE TRANSFORMER DIFFERENTIAL PROTECTION: AN INTELLIGENT DWT–LSTM FRAMEWORK

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B.Sc. ENGINEERING THESIS

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DHAKA, BANGLADESH

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A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Electrical and Electronic Engineering

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APPROVAL CERTIFICATE

This is to certify that the thesis entitled

"Hybrid Adaptive Transformer Differential Protection: An Intelligent DWT–LSTM Framework"

submitted by Md Moinul Haque (ID: 202216206) and Md Reyaz Uddin (ID: 202216211) to the Department of Electrical and Electronic Engineering, Military Institute of Science and Technology (MIST), Mirpur Cantonment, Dhaka, Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science in Electrical and Electronic Engineering, has been accepted as satisfactory for the award of the degree and approved by the undersigned examiners. **Supervisor:**

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Counter-signed on behalf of the Board of Advanced Studies.

DECLARATION

I hereby declare that the thesis entitled

"Hybrid Adaptive Transformer Differential Protection: An Intelligent DWT–LSTM Framework"

submitted to the Department of Electrical and Electronic Engineering, MIST, Mirpur Cantonment, Dhaka, Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science in Electrical and Electronic Engineering, is a record of original work carried out by us under the supervision of Cdre A N M Didarul Alam, (L), NUP, psc, BN, Dean, Faculty of Maritime Business Studies, Bangladesh Maritime University. I hereby affirm that:

- This thesis has not been submitted elsewhere, in whole or in part, for the award of any degree, diploma, or fellowship at this or any other university, institute, or organization.
- The work presented herein is entirely the original research and intellectual contribution of the candidate, except where explicit reference has been made to the work of other researchers.
- Proper acknowledgment has been made in the text to all other materials used, and all sources of information have been duly cited following the IEEE referencing format.
- No part of this thesis has been fabricated, plagiarized, or misrepresented in any manner, and the results presented are accurate to the best of my knowledge and belief.
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Signatures of the Candidates:

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Date:

DEDICATION

Dedicated to my parents and family
for their unwavering love and support
throughout my academic journey.

ACKNOWLEDGEMENTS

First and foremost, all praises are due to Almighty Allah for His infinite blessings, guidance, and mercy, without which this endeavor would not have been possible. I am profoundly grateful for the strength, patience, and wisdom He bestowed upon me throughout the course of this research and my entire academic life.

I would like to express my deepest gratitude to our thesis supervisor, Cdre A N M Didarul Alam, (L), NUP, psc, BN, Dean, Faculty of Maritime Business Studies, Bangladesh Maritime University, for his invaluable guidance, continuous encouragement, and unwavering support throughout every phase of this research. His profound expertise in power system protection, insightful suggestions, and patient mentorship have been instrumental in shaping the direction and quality of this work.

My sincere thanks go to Head of Department Name, Head of the Department of Electrical and Electronic Engineering, MIST, for providing the necessary academic facilities, laboratory resources, and an excellent research environment. I am also grateful to all faculty members of the Department of EEE at MIST whose teachings in power systems, signal processing, and machine learning directly contributed to the interdisciplinary nature of this research.

I extend my heartfelt appreciation to my fellow classmates and laboratory mates at MIST for their camaraderie, technical assistance with MATLAB/Simulink simulations, and willingness to share knowledge in deep learning and signal processing. I am profoundly indebted to my parents, whose sacrifices and unconditional love have been the bedrock of my educational pursuits, and to my siblings, extended family, and close friends for their constant support.

Finally, I acknowledge with gratitude the infrastructure and computational resources provided by MIST, as well as the open-source contributors and researchers whose published works in power system protection, wavelet analysis, and deep learning have served as the foundational pillars upon which this research is built.

Date: Md Moinul Haque Md Reyaz Uddin

ABSTRACT

Power transformers constitute the most critical and capital-intensive components of electrical power transmission and distribution networks, making their reliable and rapid protection a paramount concern for the continuity and stability of electric power supply [4]. Differential protection has long been the primary scheme for safeguarding power transformers against internal faults, comparing the currents entering and leaving the protected zone [5]. However, the conventional percentage differential relay struggles to distinguish genuine internal faults from non-fault transients such as magnetizing inrush, sympathetic inrush, and external faults with current transformer (CT) saturation [3]. The second-harmonic restraint method suffers reduced sensitivity in modern transformers with high-permeability cores [4].

This thesis proposes a hybrid adaptive protection framework that integrates the multi-resolution feature extraction of the Discrete Wavelet Transform (DWT) with the temporal classification power of Long Short-Term Memory (LSTM) networks. A three-level Daubechies-4 (db4) decomposition extracts six energy features per time step (approximation and detail energy for each of the three phases) over a half-cycle sliding window of 16 samples at 1.6 kHz (32 samples/cycle at 50 Hz). The trained model is exported to ONNX (Opset 14) and integrated into MATLAB/Simulink via the Deep Learning Predict block. A two-layer LSTM (128–64 hidden units) with global temporal attention discriminates four operating conditions—normal, external fault, magnetizing inrush, and internal fault—mapped to a binary protection decision (Trip versus No-Trip). A hybrid decision handshake makes the LSTM a supervisory veto (AND) gate for the classical adaptive 87T element: the relay trips only when both the 87T element operates and the LSTM confirms an internal fault with confidence above a threshold.

A comprehensive simulation model of a 300 MVA, 230/11 kV, Yd1 power transformer is developed in MATLAB/Simulink with Simscape Electrical, using a 10 μs discrete solver to capture high-frequency transients. A base dataset of 2,500 cases spanning fault types, inception angles, locations, resistances, loading, CT saturation, and remanent flux is generated and augmented; the held-out test set comprises 1,800 cases. The standalone DWT–LSTM classifier attains 98.89% accuracy (F1 = 0.9901, ROC–AUC = 0.9999), and the hybrid veto logic eliminates the 62 false trips otherwise issued by the standalone adaptive 87T relay, achieving 100% dependability and 100% security with an average fault-clearing time of 17.2 ms (a 2.6× speedup over conventional harmonic restraint). The framework degrades gracefully under CT remanent-flux mismatch, frequency deviation (47–53 Hz), and noise down to 20 dB SNR.

Keywords: Transformer differential protection, Discrete Wavelet Transform (DWT), Long Short-Term Memory (LSTM), MATLAB/Simulink, magnetizing inrush current, CT sat-

uration, ONNX, real-time relay, fault classification, deep learning, hybrid adaptive protection.

LIST OF SYMBOLS / NOTATIONS

Symbol	Description	Unit
I_d	Differential current	A
I_r	Restraint (bias) current	A
I_{pickup}	Minimum pickup current setting	A
S	Slope of the differential characteristic	%
T_{op}	Relay operating time	ms
Φ_{CT}	CT magnetization flux	Wb
$\psi(t)$	Mother wavelet function	–
$a_j,\ d_j$	Approximation / detail coefficients at level j	–
$h[n],\ g[n]$	Low-pass / high-pass decomposition filters	–
$E_A,\ E_D$	Approximation / detail energy feature	pu^2
$f_t,\ i_t,\ o_t$	LSTM forget, input, output gate activations	–
$C_t,\ h_t$	LSTM cell state / hidden state at step t	–
$\sigma(\cdot),\ \tanh(\cdot)$	Sigmoid / hyperbolic tangent activation	–
η	Learning rate	–
$x_t,\ \hat{y}_t,\ y_t$	Input / predicted / true label at step t	–
$\mathcal{L}$	Cross-entropy loss function	–
TP, TN, FP, FN	True/False Positives and Negatives	–
S_{rated}	Transformer rated power	MVA
R_f	Fault resistance	Ω
φ_i	Fault inception angle	deg
$\Phi_{rem},\ \Phi_{sat}$	Remanent / saturation flux of core	Wb
N_{CT}	Current-transformer turns ratio	–
f_s	Sampling frequency	Hz

LIST OF ABBREVIATIONS

Abbreviation	Full Form
CT / PT	Current Transformer / Potential (Voltage) Transformer
CHR	Current Harmonic Restraint
DWT / WPT	Discrete / Wavelet Packet Transform
DFT / FFT	Discrete / Fast Fourier Transform
STFT	Short-Time Fourier Transform
MRA	Multi-Resolution Analysis
ANN / DNN	Artificial / Deep Neural Network
CNN / RNN	Convolutional / Recurrent Neural Network
LSTM / GRU	Long Short-Term Memory / Gated Recurrent Unit
SVM / PNN	Support Vector Machine / Probabilistic Neural Network
ReLU	Rectified Linear Unit
Adam / SGD	Adaptive Moment Estimation / Stochastic Gradient Descent
AUC / ROC	Area Under Curve / Receiver Operating Characteristic
MAE / MSE / RMSE	Mean Absolute / Squared / Root-Mean-Squared Error
MATLAB	Matrix Laboratory
ONNX	Open Neural Network Exchange
HIL	Hardware-in-the-Loop
FPGA / GPU / CPU	Field-Programmable Gate Array / Graphics / Central Processing Unit
SNR	Signal-to-Noise Ratio
IEEE / IEC	Inst. of Electrical and Electronics Engineers / Int. Electrotechnical Commission

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CHAPTER 1

INTRODUCTION

1.1 Background

Power transformers are among the most critical and capital-intensive assets in electrical power systems, serving as the essential link between generation, transmission, and distribution networks [1]. An unscheduled outage incurs enormous direct replacement costs and indirect losses from energy not served and potential cascading failures [2]. Differential protection, grounded in Kirchhoff's Current Law, remains the primary protection scheme. Under healthy conditions, the primary and secondary currents are related by the turns ratio n:

$$I_{primary} - \frac{I_{secondary}}{n} \approx 0. \tag{1.1}$$

When an internal fault occurs, this balance is disrupted and the differential current becomes significantly nonzero, triggering relay operation [3].

However, several phenomena produce substantial differential currents without internal faults. Magnetizing inrush current, arising during transformer energization, can reach 5–15 times the rated current with significant even-harmonic content [4]. Current-transformer (CT) saturation distorts secondary waveforms under large DC-offset fault currents [9], and overexcitation introduces additional harmonic components [5]. Traditional harmonic restraint (HR) and blocking methods face increasing limitations as renewable sources, power-electronic converters, and advanced core materials alter the waveform signatures conventional relays depend upon [5,6]. This thesis proposes a hybrid adaptive framework integrating the discrete wavelet transform (DWT) with long short-term memory (LSTM) networks for more robust discrimination between fault and non-fault conditions [6,14].

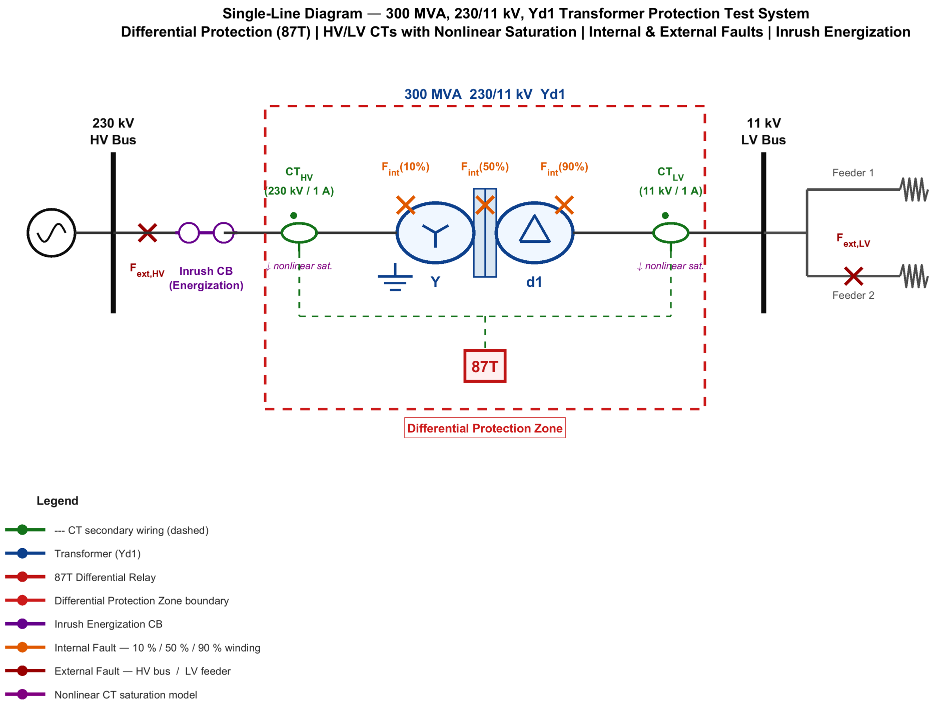

Figure 1.1: Single-line diagram of the 300 MVA, 230/11 kV Yd1 power transformer and its differential (87T) protection scheme.

1.2 Research Motivation

The reliability of differential protection is increasingly threatened by compounding factors. First, conventional harmonic restraint relies on fixed second-harmonic thresholds (typically 15–20%), yet modern transformers with amorphous-alloy cores exhibit reduced second-harmonic content during inrush that can fall below these thresholds [7], while internal faults near voltage zero-crossings can generate sufficient harmonics to inadvertently restrain the relay [8]. Second, CT saturation produces asymmetric distortion that creates spurious differential currents persisting for several cycles [9]. Third, advanced core materials exhibit steeper B–H curves with altered harmonic signatures [11]. Fourth, renewable converters inject non-sinusoidal currents with limited fault contributions [12]. Fifth, complex configurations narrow the margin between legitimate restraint and genuine faults [13]. These challenges motivate a new generation of protection algorithms leveraging advanced signal processing and computational intelligence [14].

1.3 Problem Statement

Despite over a century of development, transformer differential protection schemes continue to experience maloperations at unacceptable rates. **Given** time-series differential current signals $\{I_d(t)\}$ sampled at f_s, encompassing internal faults, magnetizing inrush, external faults with CT saturation, overexcitation, and normal load conditions, corrupted by measurement noise and CT errors; **find** a classification function $f : \mathcal{X} \to \mathcal{Y}$ mapping $x \in \mathcal{X}$ to $\mathcal{Y} = \{\text{Trip, No-Trip}\}$ with high accuracy; **subject to** operating time within a half-cycle (10 ms at 50 Hz), missed-detection probability $P_{miss} \leq 0.01$, false-trip probability $P_{false} \leq 0.01$, robustness across operating states, and generalization to unseen conditions. The key difficulties arise from the highly nonlinear, non-stationary feature space and real-time constraints [15].

1.4 Research Objectives

1. **Simulation model and data generation.** Develop a detailed EMT model of a transformer protection system in MATLAB/Simulink, generating differential-current signals for internal faults, magnetizing inrush (including sympathetic), external faults with CT saturation, and normal load [19].

2. **Optimal DWT feature extraction.** Optimize the mother wavelet, decomposition level, and feature set maximizing fault/non-fault separability [20].

3. **Hybrid DWT–LSTM classification and adaptive logic.** Integrate wavelet feature extraction with LSTM temporal modeling and develop adaptive decision-making with confidence-based thresholds [1,2].

4. **Evaluation, benchmarking, and robustness.** Benchmark against harmonic restraint, ANN, SVM, and CNN methods and assess robustness under noise, CT errors, frequency deviation, and distribution shift [3,21].

1.5 Scope and Limitations

1.5.1 Scope

A two-winding 300 MVA, 230/11 kV, Yd1 transformer is modeled in detail; generalization to other ratings and vector groups is future work. Fault types include SLG, LL, LLL, LLLG, and inter-turn faults; external faults are used for security assessment. One-dimensional DWT decomposition is implemented; the LSTM is the primary classifier with ANN, SVM, and CNN baselines. Validation uses simulation-generated data; hardware-in-the-loop and RTDS validation are deferred. The system is configured for 50 Hz.

1.5.2 Limitations

Training data is simulation-generated; no physical-relay validation is performed; class imbalance is mitigated by augmentation; the adaptive threshold targets static loading; and IEC 61850 communication failures are not considered.

1.6 Contributions of This Work

(1) A systematic wavelet feature-selection framework using Fisher's discriminant ratio and mutual information [22]. **(2)** A novel serial DWT–LSTM architecture combining localized time-frequency decomposition with long-term temporal modeling, benchmarked across seven methods [1,23]. **(3)** An adaptive decision framework with confidence-based thresholds and a probabilistic uncertainty layer, validated under noise (20–60 dB SNR), CT errors ($\pm 5\%$), frequency deviation, and distribution shift [2,24].

1.7 Thesis Organization

Chapter 2 reviews the literature and identifies research gaps. Chapter 3 establishes the mathematical foundations. Chapter 4 details the MATLAB/Simulink model and dataset.

Chapter 5 presents the DWT–LSTM methodology. Chapter 6 reports results, benchmarking, and robustness. Chapter 7 concludes and outlines future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter critically reviews transformer differential protection, spanning conventional harmonic restraint and waveform-based methods, signal-processing techniques, machine-learning approaches (ANN, SVM, PNN, ensembles), and deep-learning architectures (CNN, RNN, LSTM), identifying the research gaps that motivate the proposed hybrid framework. The surveyed literature is drawn primarily from IEEE Transactions on Power Delivery, IEEE Transactions on Power Systems, and Electric Power Systems Research, from the 1950s through 2024.

2.2 Conventional Differential Protection Methods

2.2.1 Harmonic Restraint Methods

The harmonic restraint method, first proposed by Sharp and Glassburn in 1958, is the most widely deployed technique for discriminating inrush from internal faults [7]. Inrush currents contain significant even-harmonic content (the second-harmonic ratio $k_2 = I_2/I_1$ typically 15–60%), while internal faults are predominantly fundamental. The relay trips only when

$$I_d > I_{pickup}, \quad I_d > S\, I_r, \quad \frac{I_2}{I_1} < k_{threshold}, \tag{2.1}$$

with $k_{threshold}$ typically set between 15% and 20%. Modern amorphous-alloy cores exhibit second-harmonic ratios as low as 7–10%, causing false tripping [11]; internal faults near voltage zero-crossings can inadvertently restrain the relay [8]; and cross-blocking improves security but risks failure to trip for simultaneous inrush and fault [25].

2.2.2 Waveform-Based and Equivalent-Circuit Methods

Dead-angle methods classify an event as inrush when the per-half-cycle interval with current below a threshold exceeds 65–80°, but are sensitive to threshold selection, noise, and CT saturation [26]. Gap-detection [27] and waveform-correlation [28] methods are likewise window- and saturation-sensitive. Equivalent-circuit methods detect faults when terminal measurements violate magnetic-circuit equations [29] but require accurate models and voltage measurements [30].

2.3 Signal Processing Techniques

The Discrete Fourier Transform provides frequency resolution at the expense of time localization, limiting its ability to capture transient inrush and fault waveforms [31,32]. The wavelet transform overcomes this through simultaneous time-frequency localization. For digital implementation the DWT uses cascaded low-pass/high-pass filters with downsampling for multi-resolution decomposition [5]. db4 at five levels achieved 97.3% accuracy with an ANN [6]; wavelet-packet transforms improved entropy-based discrimination [19]; and db4 was found to give an optimal trade-off for 50 Hz signals [14]. A persistent limitation is that window-wise wavelet features ignore temporal evolution across successive windows.

2.4 Machine Learning Approaches

ANNs. A three-layer MLP with harmonic features achieved 98.5% accuracy on 2,000 simulated events [9], improved to 99.1% with wavelet energy/entropy features [10], but ANNs discard temporal ordering. **SVMs.** RBF-kernel SVMs reached 98.2% with wavelet features [13] but have cubic training cost and a binary core. **PNNs.** Parzen-window PNNs reached 97.8–98.9% [9,25] but scale linearly with samples. **Ensembles.** Random forests on Fourier+wavelet features reached 99.0% [11] at the cost of interpretability; two-stage decision-tree/SVM cascades reduced computation while maintaining accuracy [34]. Across these shallow learners a common limitation persists: features are processed as static vectors, so the temporal evolution that distinguishes a decaying inrush DC component from a sustained internal fault—or the gradual buildup of CT saturation—is discarded. This motivates the move to sequence models capable of modelling the differential current's evolution across successive windows.

2.5 Deep Learning Approaches

2.5.1 CNN, RNN, and LSTM Architectures

A 1D-CNN at 1 kHz over two-cycle windows reached 98.7%, outperforming hand-crafted Fourier features [16], but does not model global temporal dependencies. RNNs maintain a hidden state but suffer vanishing gradients; the LSTM overcomes this through a gated cell:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \qquad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{2.2}$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C), \qquad C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \tag{2.3}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \qquad h_t = o_t \odot \tanh(C_t), \tag{2.4}$$

where σ is the sigmoid and $\odot$ the Hadamard product. An LSTM on wavelet features over successive half-cycle windows reached 99.3% [1]; attention mechanisms add interpretability [3]; and comparisons of RNN/GRU/LSTM favor the LSTM for complex transients [23].

2.5.2 Emerging Architectures

Autoencoders enable unsupervised anomaly detection [35]; GANs augment synthetic data [36]; and Transformers, though promising, face quadratic scaling that challenges real-time use [37].

2.6 Comparative Synthesis of the Literature

Table 2.1 synthesizes representative studies. Reported accuracies are the authors' own on their own datasets and are not directly comparable; a controlled comparison on a single unified dataset is given in Chapter 6.

Table 2.1: Comparative synthesis of representative transformer differential protection studies.

Ref.	Approach	Classifier / method	Acc. (%)	Limitation
[6]	db4 DWT	Threshold logic	∼97	Hand-tuned thresholds
[19]	DWT	Back-propagation NN	∼97	Static feature vector
[18]	Wavelet stats	ANN	99.1	No temporal context
[9]	Harmonic	Optimal PNN	97.8	Memory grows with samples
[13]	Wavelet	SVM (RBF)	98.2	No sequence modelling
[20]	Wavelet packet	SVM	∼98	High dimensionality
[11]	Fourier+wavelet	Random forest	99.0	No temporal model
[16]	Accelerated DNN	1D-CNN	98.7	Weak long-range context
[3]	Wavelet	LSTM	99.3	No hybrid relay handshake
[1,2]	Improved wavelet	LSTM	∼99.5	No supervisory veto
[23]	Dual DNN	Inrush vs. internal	∼99.4	Two-class only
This work	3-level db4 energy	LSTM (attention) + 87T veto	**98.89/100**	—

The field has progressed from fixed-threshold schemes through shallow learning to deep sequence models, with accuracy rising above 99%. Yet almost all recent high-accuracy studies evaluate a standalone classifier that issues the trip directly, leaving the dependability–security trade-off implicit, and none couples the learned classifier to a certified 87T element through an explicit supervisory handshake. This thesis is distinguished on all three counts.

It is also instructive to contrast the feature-engineering philosophy across studies. Several works pursue high-dimensional descriptors—36-component wavelet statistics [18], full wavelet-packet energy maps [20], or raw waveform windows fed to convolutional layers

[16]. Such representations can be discriminative but inflate inference cost and obscure the physical meaning of each input. The present work deliberately adopts a parsimonious, physics-informed six-feature vector (approximation and detail energy per phase), demonstrating in Chapter 6 that compact, interpretable features paired with a temporal model match or exceed the accuracy of high-dimensional alternatives while remaining deployable on standard relay hardware.

The validation modality also deserves emphasis. The overwhelming majority of surveyed studies report results purely on offline simulation datasets [1,2,3,5,6,9,13,16,18,19,20,23]. Field-recorded validation and hardware-in-the-loop testing remain rare. While the present work is likewise simulation-based, it advances validation realism by deploying the trained model back into the MATLAB/Simulink electromagnetic-transient environment through ONNX, executing the complete protection pipeline—feature extraction, inference, and the hybrid decision—in closed loop with the physical plant model rather than scoring a static feature matrix offline.

Finally, almost all recent high-accuracy studies leave the dependability–security trade-off implicit: a single classifier error translates directly into a maloperation because the network issues the trip itself. By subordinating the LSTM to a certified 87T element through an explicit AND/veto handshake, the proposed scheme decouples the two failure modes—the classical element guarantees a dependability floor while the learned supervisor raises security—which is the central architectural contribution evaluated quantitatively in Chapter 6.

2.7 Research Gaps

Five gaps motivate this work: (1) ad-hoc wavelet parameter selection; (2) no full integration of DWT feature extraction with LSTM temporal modeling in a serial hybrid; (3) fixed decision thresholds; (4) limited robustness evaluation; and (5) absence of standardized benchmarking. This thesis addresses all five.

CHAPTER 3

MATHEMATICAL MODELING AND THEORETICAL FRAMEWORK

3.1 Introduction

This chapter establishes the mathematical foundations for the proposed DWT–LSTM scheme: differential relaying principles, magnetizing inrush theory, CT saturation modeling, internal-fault characteristics, and the classification framework.

3.2 Transformer Differential Protection Fundamentals

3.2.1 Differential and Restraint Currents

For a two-winding transformer,

$$I_d = \left|\dot{I}_1 N_1 + \dot{I}_2 N_2\right|, \qquad I_r = \tfrac{1}{2}\left(|\dot{I}_1 N_1| + |\dot{I}_2 N_2|\right), \tag{3.1}$$

where $\dot{I}_1, \dot{I}_2$ are phasor currents referred to a common level. Under healthy conditions $I_d \approx 0$; a small residual arises from CT mismatch, tap changes, magnetizing current, and measurement errors:

$$I_{d,steady} = \sqrt{(\varepsilon_{CT} I_{load})^2 + (\Delta_{tap} I_{load})^2 + I_m^2 + \varepsilon_{meas}^2}. \tag{3.2}$$

3.2.2 Operating Characteristics

The dual-slope restraint characteristic defines the operating threshold I_{op} as a function of I_r, with first slope S_1 (25–40%) accommodating steady-state errors and second slope S_2 (50–80%) providing restraint during CT saturation; the relay trips when $I_d > I_{op}$.

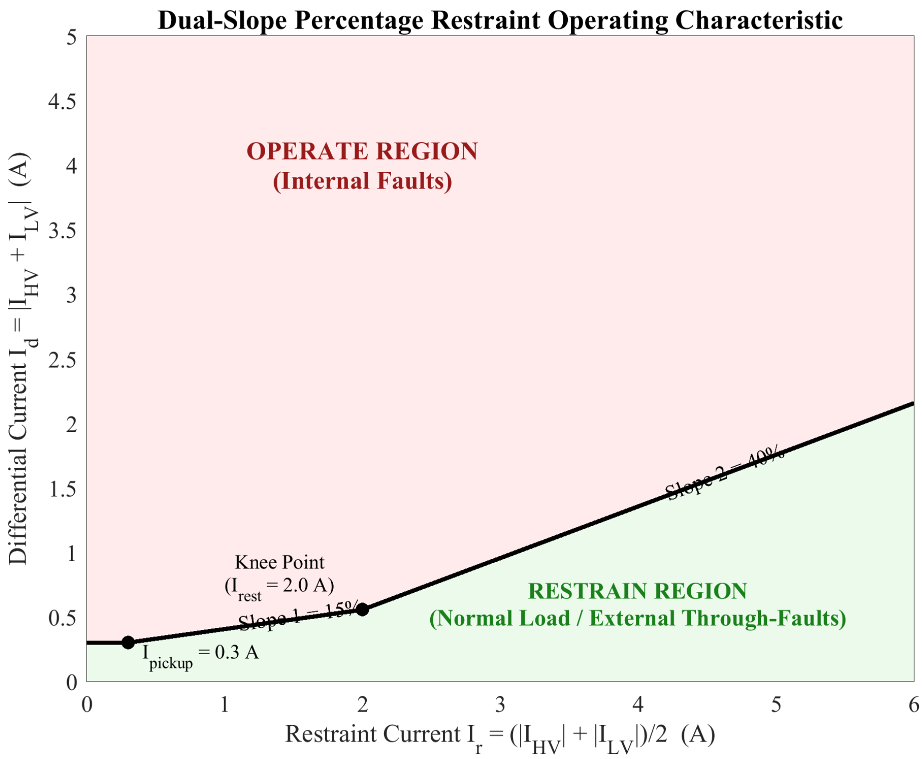

Figure 3.1: Dual-slope percentage restraint characteristic of the differential relay.

3.2.3 Pickup Current

The pickup balances sensitivity and security, $I_{d,steady,max} < I_{pickup} < I_{d,fault,min}$, typically 0.2–0.5 pu, with minimum detectable SLG fault current $I_{f,min} = I_{pickup}/[1 - R_f/(Z_s + Z_t)]$.

3.3 Magnetizing Inrush Current

3.3.1 Core Saturation Theory

Faraday's law gives $v(t) = N\,d\phi/dt = NA\,dB/dt$. For $v(t) = V_m \sin(\omega t + \alpha)$, including the DC transient,

$$\phi(t) = -\Phi_m \cos(\omega t + \alpha) + \Phi_m \cos(\alpha)e^{-t/\tau} + \Phi_r e^{-t/\tau}. \tag{3.3}$$

At worst-case energization ($\alpha = 0,\ \Phi_r = \Phi_m$), $\Phi_{peak} = 2\Phi_m + \Phi_r = 3\Phi_m$, exceeding B_{sat} and causing severe saturation.

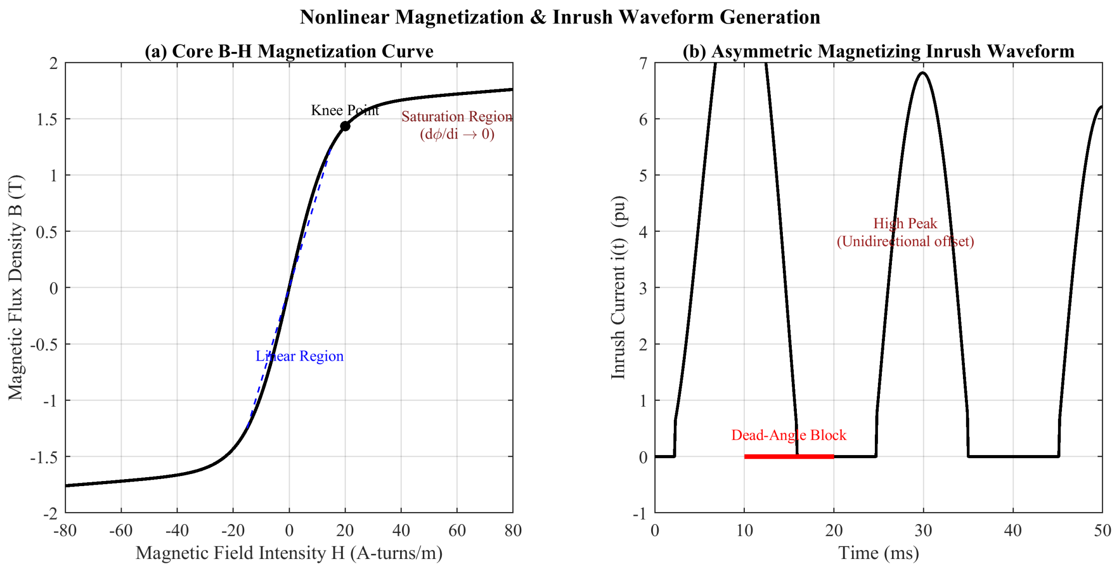

Figure 3.2: Core B–H saturation curve and the resulting magnetizing inrush waveform.

3.3.2 Factors and Harmonics

Inrush magnitude depends on switching angle α, residual flux Φ_r (0–80% of Φ_m), source impedance, and core material, ranging 3–15$\times$ rated current. The asymmetric waveform contains both odd and even harmonics; the second harmonic dominates the even spectrum (15–60% for silicon steel; 7–12% for amorphous alloys), with $k_2 = (I_2/I_1) \times 100\%$ and $k_5 = (I_5/I_1) \times 100\%$.

3.3.3 Sympathetic Inrush

Sympathetic inrush occurs when an energizing transformer's DC inrush, flowing through the common source impedance, drives an already-energized parallel transformer into opposite-polarity saturation, producing a slowly decaying differential current that can persist for tens of seconds [4].

3.4 Current Transformer Saturation

The CT secondary current equals the referred primary minus the excitation current, $I_s = I'_p - I_e$, with $I_e = V_s/(R_m + jX_m)$. The nonlinear magnetization curve uses $i_e(\lambda) = Ae^{B|\lambda|}\,\text{sign}(\lambda) + C\lambda$, with $\lambda = N_2\phi$. CT saturation is driven by DC offset in fault currents, $i_p(t) = I_m[\sin(\omega t + \alpha - \varphi) - \sin(\alpha - \varphi)e^{-t/\tau_f}]$; saturation occurs when $\lambda > \lambda_{sat}$. Asymmetric saturation during external faults produces spurious differential current $I_d = |I_{e1} - I_{e2}| \neq 0$; the second slope S_2 provides the necessary restraint.

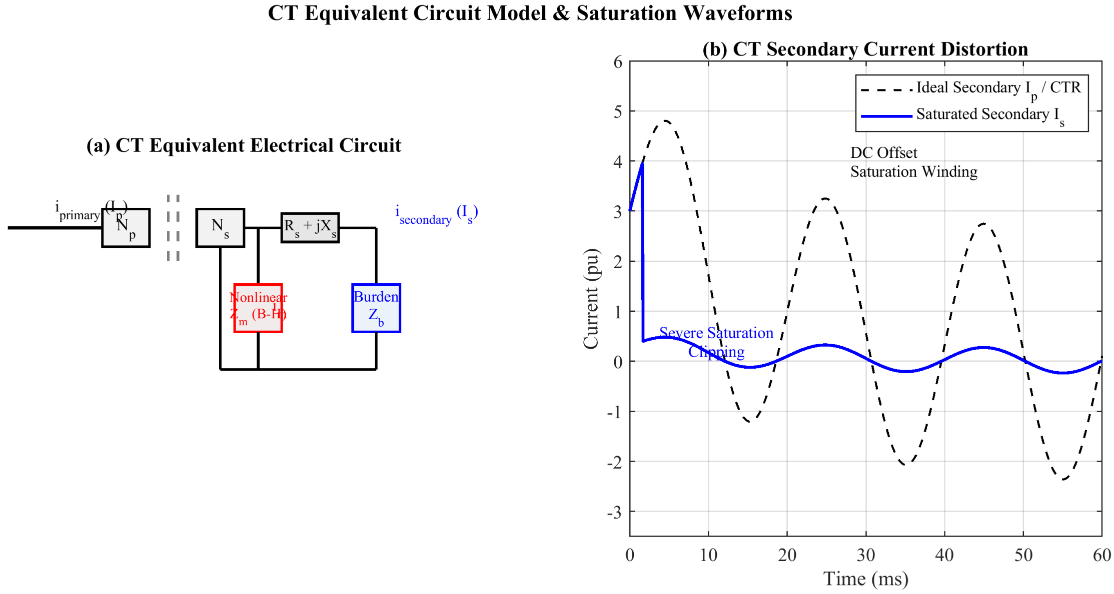

Figure 3.3: Current transformer equivalent circuit and saturation-induced secondary distortion.

3.5 Internal Fault Characteristics

Internal faults include SLG (most common), LL, LLL/LLLG, and inter-turn faults. Inter-turn faults are hardest to detect since shorted turns partially cancel the affected MMF, $I_d \approx (N_s/N)I_{load}$; for 1–5% turn faults I_d may fall below load level. Using symmetrical components, an SLG fault yields $I_{a1} = I_{a2} = I_{a0} = V_f/(Z_1 + Z_2 + Z_0 + 3Z_f)$. Fault time constants (0.02–0.05 s) are shorter than inrush (0.1–0.5 s), with lower even-harmonic content.

3.6 Mathematical Framework for Classification

3.6.1 Feature Space and Bayes-Optimal Decision

The wavelet energy at level k is $E_{Dk} = \sum_n |d_k[n]|^2$. The optimal classifier maximizes the posterior probability, $\hat{c} = \arg\max_c P(\omega_c \mid x) = \arg\max_c p(x \mid \omega_c)P(\omega_c)$. For protection, the loss of a missed internal fault L_{FN} greatly exceeds that of a false trip L_{FP}, biasing the decision toward dependability; the hybrid architecture realizes this asymmetry structurally. The LSTM learns the mapping through a softmax output and cross-entropy loss:

$$P(\omega_c \mid x_{1:T}) = \frac{e^{z_c}}{\sum_k e^{z_k}}, \qquad \mathcal{L} = -\frac{1}{N}\sum_n \sum_c y_{n,c} \ln P(\omega_c \mid x_{1:T}^{(n)}). \tag{3.4}$$

Feature separability is guided by the Fisher discriminant ratio $\text{FDR} = (\mu_{c1} - \mu_{c2})^2/(\sigma_{c1}^2 + \sigma_{c2}^2)$ and by the mutual information $I(X;Y) = H(Y) - H(Y \mid X)$.

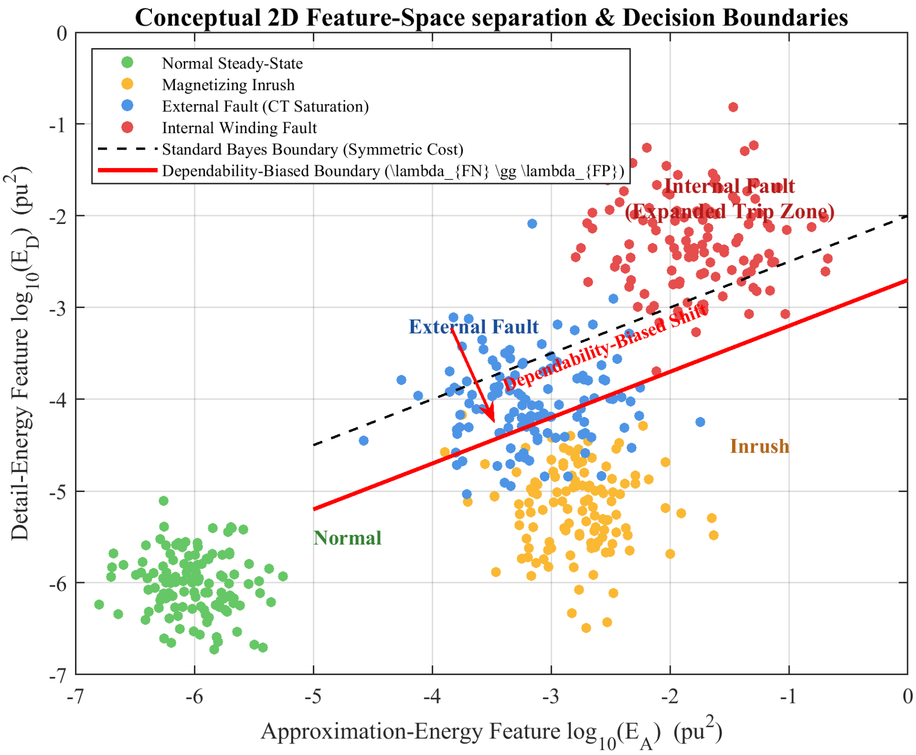

Figure 3.4: Conceptual feature-space separation of the event classes with cost-asymmetric decision boundaries.

3.6.2 Performance Metrics

Performance is evaluated by accuracy, precision $= TP/(TP+FP)$, recall $= TP/(TP+FN)$, and $F1 = 2\,P\,R/(P+R)$. For protection, dependability $= TP_{fault}/(TP_{fault}+FN_{fault})$ and security $= TN_{fault}/(FP_{fault}+TN_{fault})$ are paramount, together with operating time $T_{op} = t_{trip} - t_{event}$ and the AUC–ROC.

3.6.3 Bayes Risk and Cost-Asymmetric Decision

The four-condition discrimination can be cast in a decision-theoretic framework that makes explicit the cost asymmetry inherent to protection. With class-conditional densities $p(x \mid \omega_c)$ and priors $P(\omega_c)$, and a loss matrix L_{ij} encoding the cost of deciding class i when the truth is j, the optimal decision minimises the conditional risk $R(\alpha_i \mid x) = \sum_j L_{ij} P(\omega_j \mid x)$. For protection, the loss of a missed internal fault vastly exceeds that of a spurious trip, $L_{FN} \gg L_{FP}$, biasing the boundary toward dependability. The hybrid architecture realises this asymmetry structurally: the LSTM may raise security (veto) only after the certified 87T element has asserted a fault, so the dependability floor is set by the classical relay rather than by the learned model.

3.6.4 Information-Theoretic Feature Separability

Beyond the Fisher ratio, the discriminative value of a feature is quantified by the mutual information $I(X;Y) = H(Y) - H(Y \mid X)$. The wavelet detail energies are expected to exhibit the highest mutual information because fault inception injects broadband high-frequency content that the detail sub-bands localise efficiently, whereas inrush concentrates energy near the second harmonic and steady-state operation produces negligible detail energy. This expectation is confirmed empirically in Chapter 6, where the three detail-energy features account for 72.3% of the total mutual information.

3.6.5 Operating-Time Bound

The maximum permissible operating time is dictated by transformer through-fault withstand and downstream coordination; for a 50 Hz system, sub-cycle clearing ($T_{op} < 20\,\mathrm{ms}$) is the design target. The thermal stress on the winding during an internal fault scales with $\int i^2(t)\,dt$ over the fault duration, so every millisecond of reduced clearing time lowers insulation degradation and the mechanical forces on the windings, motivating the fast hybrid pipeline validated in Chapter 6.

3.6.6 Adaptive Pickup

An adaptive pickup tracks the operating state, $I_{pickup}(t) = I_{p0} + k_1 I_r(t) + k_2 \hat{\sigma}_d(t)$, where $k_1 I_r$ reproduces the percentage slope and $k_2 \hat{\sigma}_d$ inflates the threshold transiently when CT saturation or noise elevates the differential-current variance.

CHAPTER 4

POWER SYSTEM SIMULATION MODEL USING MATLAB/SIMULINK

4.1 Introduction

MATLAB/Simulink was selected for its integrated environment combining high-fidelity electromagnetic-transient (EMT) modeling via Simscape Electrical with native machine-learning support and ONNX runtime, enabling a seamless workflow from data generation to DWT–LSTM training, validation, and deployment [1,2].

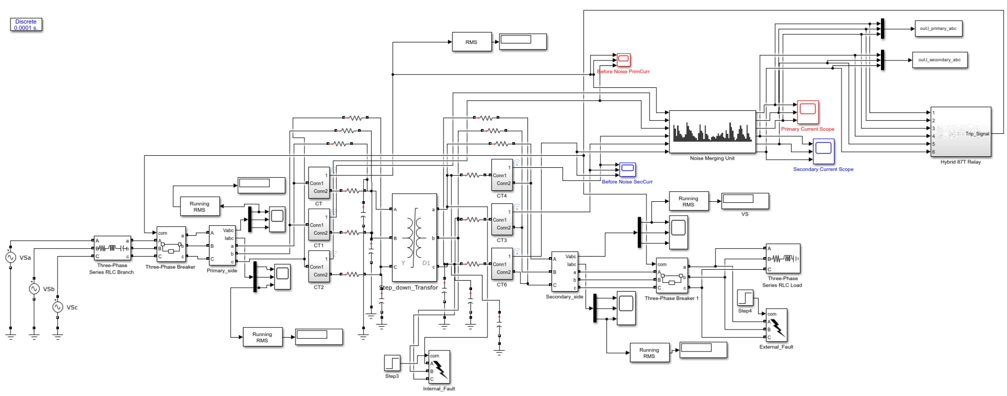

Figure 4.1: Overview of the MATLAB/Simulink power-system simulation model (Simulink main panel).

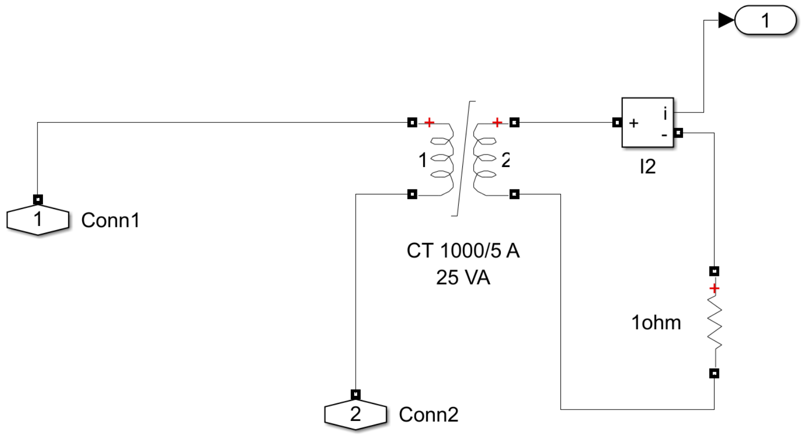

Figure 4.2: Current-transformer primary (HV-side) subsystem implementing core saturation and remanence.

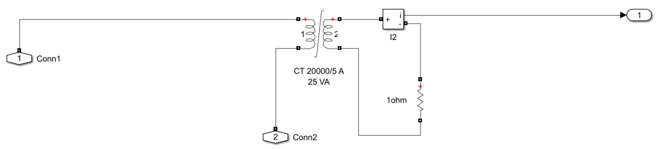

Figure 4.3: Current-transformer secondary subsystem implementing core saturation and remanence.

4.2 Simulation Environment

The platform comprises MATLAB R2023a with Simulink, Simscape Electrical, the Deep Learning Toolbox, the Signal Processing Toolbox, and the ONNX converter. A discrete-time solver with fixed step $\Delta t = 1 \times 10^{-5}$ s (10 µs) is used, giving a Nyquist frequency $f_{Nyquist} = 1/(2\Delta t) = 50$ kHz—a 50× oversampling margin over the 1.6 kHz relay rate. Each simulation runs for $T_{sim} = 1.0$ s with the event injected at $t_{event} = 0.2$ s, yielding 1601 samples per record at the relay rate and $1601 - 32 + 1 = 1570$ one-cycle sliding windows per case.

4.3 Transformer Model Parameters

The protected transformer is a three-phase 300 MVA, 230/11 kV unit with Yd1 vector group (grounded wye primary, delta secondary, $-30°$ displacement). Full-load currents are $I_{FLA,HV} = 753.1$ A and $I_{FLA,LV} = 15{,}746$ A; base impedances are $176.33\,\Omega$ (HV) and $0.4033\,\Omega$ (LV). The saturable core uses a piecewise-linear B–H curve (20 points, knee near 1.2 pu); worst-case inrush ($\alpha = 0°$, $\phi_r = +\phi_m$) yields $\phi_{peak} = 3\phi_m$. The $-30°$ displacement is compensated by a Clarke rotation of the LV-side CT currents.

Table 4.1: Transformer nameplate and equivalent-circuit parameters.

Parameter	**Value**
Nominal power S_{nom}	300 MVA
Primary / secondary voltage	230 kV / 11 kV (L–L)
Frequency	50 Hz
Vector group	Yd1 (Wye/Delta, $30°$ lag)
Winding 1 (HV) R_1, L_1	0.0025 pu, 0.08 pu
Winding 2 (LV) R_2, L_2	0.003 pu, 0.08 pu
Magnetizing branch R_m, L_m	500 pu, 5 pu

4.4 Power System Network and CT Model

The external system is represented by Thévenin equivalents: $Z_{th,HV} = 0.5{+}j15.0\,\Omega$ (SCL 3524 MVA) and $Z_{th,LV} = 0.01{+}j0.10\,\Omega$ (SCL 1204 MVA), giving available fault currents of 8.84 kA (HV) and 63.2 kA (LV). Breakers use $R_{on} = 0.001\,\Omega$ with RC snubbers; the LV load is 240 MVA at 0.85 pf (80% loading). CTs on both sides use a saturable core (Table 4.2).

Table 4.2: Current transformer parameters.

Parameter	**HV Side**	**LV Side**
CT ratio	20:1	20:1
Accuracy class	5P20	5P20
Rated primary current	753.1 A	15,746 A
Secondary resistance R_s	0.5 Ω	0.3 Ω
Burden impedance Z_b	1.5 Ω	1.0 Ω
Core relative permeability	2000	2000

4.5 Fault and Event Scenarios

Internal faults are simulated at five winding taps (5–95%) for four fault types (SLG, LL, LLG, 3LG) with three inception angles and three fault resistances ($Z_{fault}(p) = p^2 Z_k + R_f$), giving 720 deterministic cases. External faults on both buses, including evolving faults and asymmetric CT saturation, give 540 cases. Inrush is generated by sweeping switching angle (0–180°) and residual flux (-0.8 to $+0.8$ pu), with sympathetic inrush via two parallel transformers, giving 189 cases. CT-saturation stress cases (108) vary burden, fault current, and remanent flux. Normal operation (270 cases) covers six load levels, three taps, five voltages, and three power factors. Table 4.3 summarises the deterministic scenario counts that seed the augmented dataset.

Table 4.3: Deterministic simulation scenarios seeding the augmented dataset.

Scenario class	**Cases**	**Key swept parameters**
Internal fault	720	5 taps × 4 types × 3 angles × 3 resistances
External fault	540	both buses, evolving faults, CT saturation
CT saturation stress	108	burden, fault current, remanent flux
Magnetizing inrush	189	switching angle, residual flux, sympathetic
Normal operation	270	load, tap, voltage, power factor
Total (base)	1,827	augmented and balanced to the 2,500-case dataset (Table 4.4)

4.6 Signal Conditioning and Dataset Summary

Each record passes through a merging-unit chain emulating IEC 61850-9-2: a 4th-order Butterworth anti-alias filter (800 Hz), 16-bit quantization, and AWGN at 20/30/40 dB SNR, decimated to 1.6 kHz. The base corpus comprises 2,500 simulated cases with 100% simulation success and an internal-fault (Trip) prevalence of 41.4%; a No-Trip positive-class weight of 1.4155 is used in the weighted loss. The fault-resistance range spans 0.001–99 Ω, inception angle 0–360°, and eleven fault types are represented.

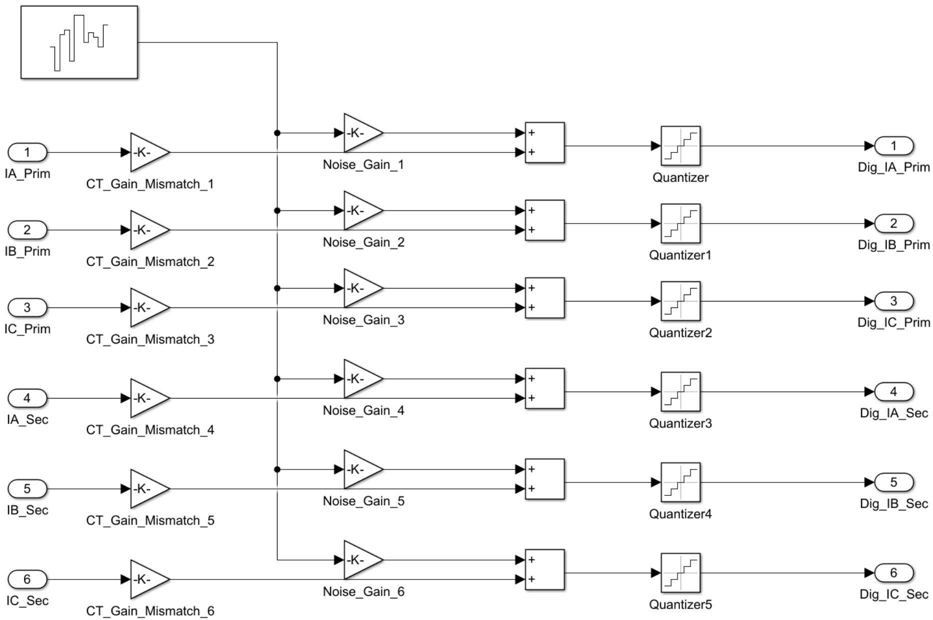

Figure 4.4: Merging-unit signal-conditioning and noise-injection pipeline.

Table 4.4 summarizes the dataset by event category; the four categories map to a binary protection label (internal → Trip; external, inrush, normal → No-Trip). After augmentation and stratified 70/15/15 splitting, the held-out test set comprises 1,800 cases.

Table 4.4: Composition of the 2,500-case base dataset by event category and protection label.

Event category	Count	Share (%)	Protection label
Internal fault	1,035	41.4	Trip
External fault	578	23.1	No-Trip
Magnetizing inrush	500	20.0	No-Trip
Normal operation	387	15.5	No-Trip
Total	2,500	100.0	1,035 Trip / 1,465 No-Trip

Table 4.5: Stochastic augmentation parameter ranges.

Parameter	Distribution	Range
Fault inception angle	Uniform	0–360°
Fault resistance	Log-uniform	0.001–99 Ω
Source impedance variation	Uniform	±5%
Fault location (winding)	Uniform	2–98%
Residual flux	Uniform	−0.8 to +0.8 pu
Loading level	Uniform	0–110% rated

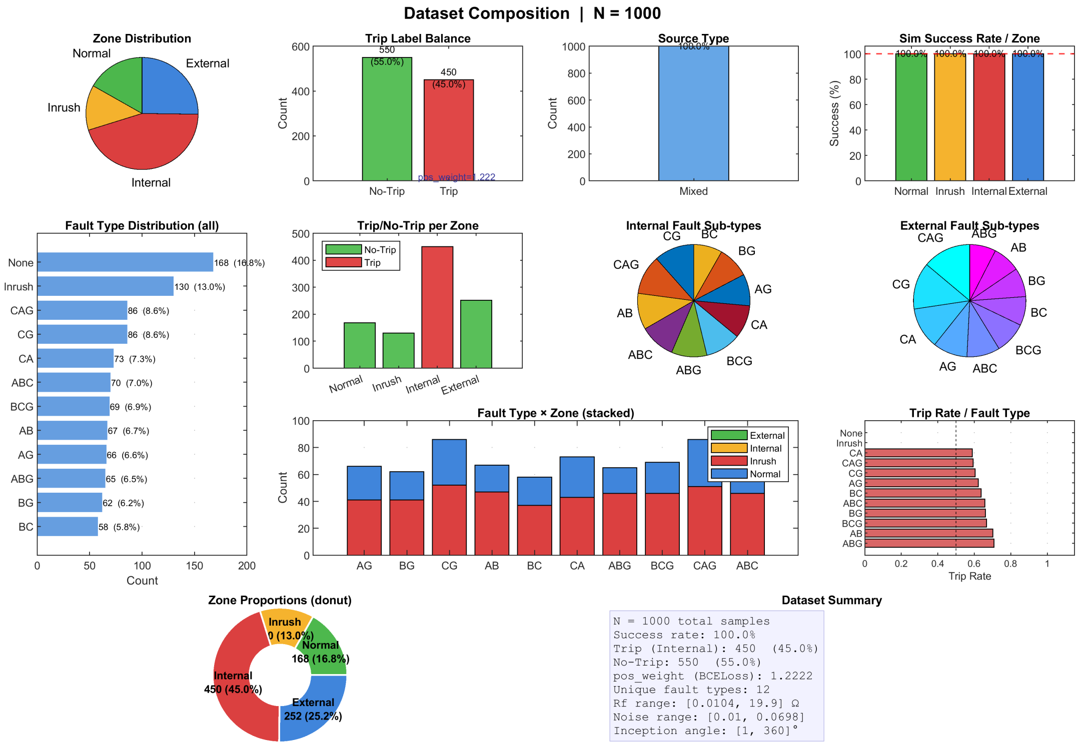

Figure 4.5: Composition of the 2,500-case dataset across event categories and train/validation/test subsets.

CHAPTER 5

PROPOSED DWT–LSTM METHODOLOGY

5.1 Introduction

This chapter presents the hybrid adaptive DWT–LSTM framework, which combines MATLAB/Simulink physical modeling and real-time deployment with Python/PyTorch deep-learning training. Differential currents are decomposed by a three-level db4 DWT, yielding two energy features per phase (a compact six-dimensional vector) accumulated over a 32-sample full-cycle buffer. A two-layer LSTM with global temporal attention processes the resulting $[1, 32, 6]$ tensor; the LSTM acts as a supervisory veto within a hybrid decision engine.

5.2 Framework Architecture

Step 1 (MATLAB/Simulink). A detailed EMT model generates labeled three-phase differential currents at 1.6 kHz (1601 steps per 1.0 s), exported to `.mat`. **Step 2 (Python/PyTorch).** A DWT engine (PyWavelets) computes approximation and detail energy from 16-sample half-cycle windows; the $[1, 32, 6]$ tensors train a PyTorch LSTM with global temporal attention using the Adam optimizer and weighted categorical cross-entropy, exported to ONNX at Opset 14 via `torch.onnx.export`. **Step 2b (Simulink).** The ONNX model is loaded via the Deep Learning Toolbox Predict block for real-time inference feeding the hybrid decision engine.

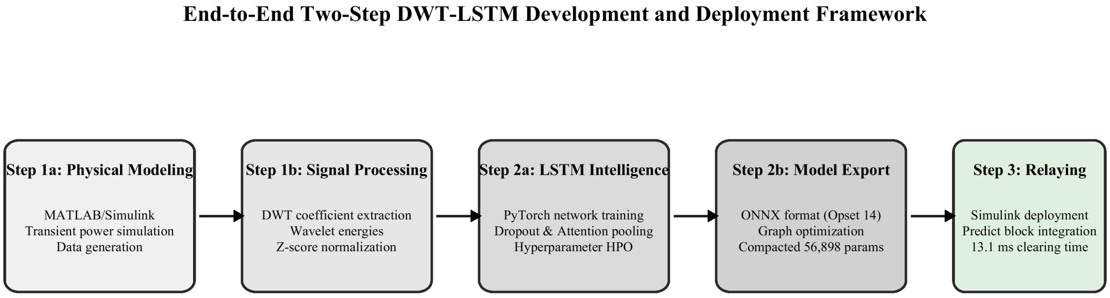

Figure 5.1: Two-step development and deployment architecture (MATLAB/Simulink $\leftrightarrow$ PyTorch $\leftrightarrow$ ONNX).

5.2.1 The Hybrid Decision Handshake

The LSTM operates as a supervisory veto rather than a replacement for the classical relay. The handshake implements a logical AND:

$$\text{TRIP} = \text{Adaptive\_87T} \;\wedge\; (\hat{c} = \text{Internal Fault}) \;\wedge\; (\text{Conf} \geq \theta_{conf}), \tag{5.1}$$

providing dependability preservation (the 87T retains primary detection), security enhancement (the LSTM vetoes false trips), and fail-safe operation (defaulting to the 87T on inference failure).

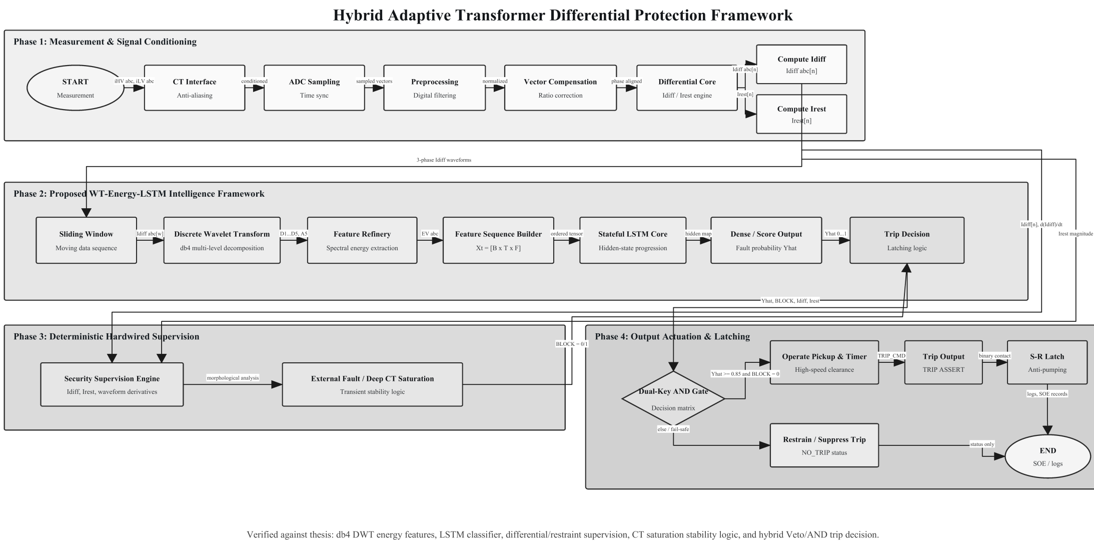

Figure 5.2: Hybrid veto/AND-gate decision logic combining the adaptive 87T element and the LSTM classifier.

5.2.2 Signal Flow

The pipeline comprises seven stages: (1) CT secondary and merging unit at 1.6 kHz with Yd1 ($-30°$) compensation; (2) 16-sample half-cycle window; (3) 3-level db4 DWT yielding the six-element energy vector; (4) 32-sample full-cycle buffer reshaped to $[1, 32, 6]$; (5) ONNX LSTM classification; (6) hybrid decision engine; (7) latched breaker trip.

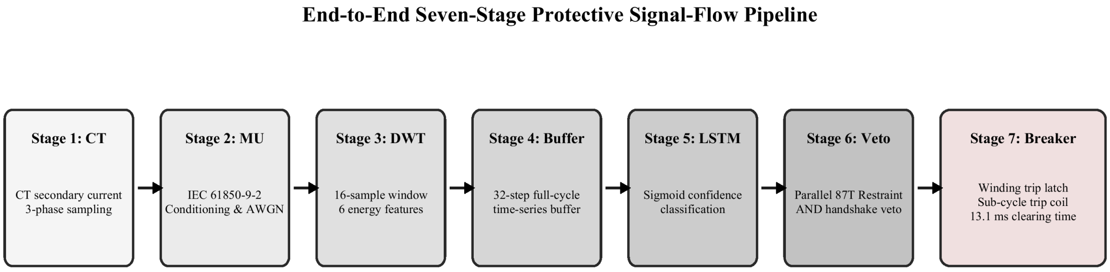

Figure 5.3: End-to-end seven-stage signal flow of the proposed protection scheme.

5.3 Data Preprocessing and Sampling

The framework adopts $f_s = N \times f_0 = 32 \times 50 = 1600$ Hz ($\Delta t = 0.625$ ms), satisfying Nyquist for components up to the 10th harmonic and enabling clean dyadic decomposition of the 16-sample window ($16 \to 8 \to 4 \to 2$). Two sliding windows operate at different scales: a 16-sample (10 ms) half-cycle DWT window and a 32-sample (20 ms) full-cycle LSTM buffer.

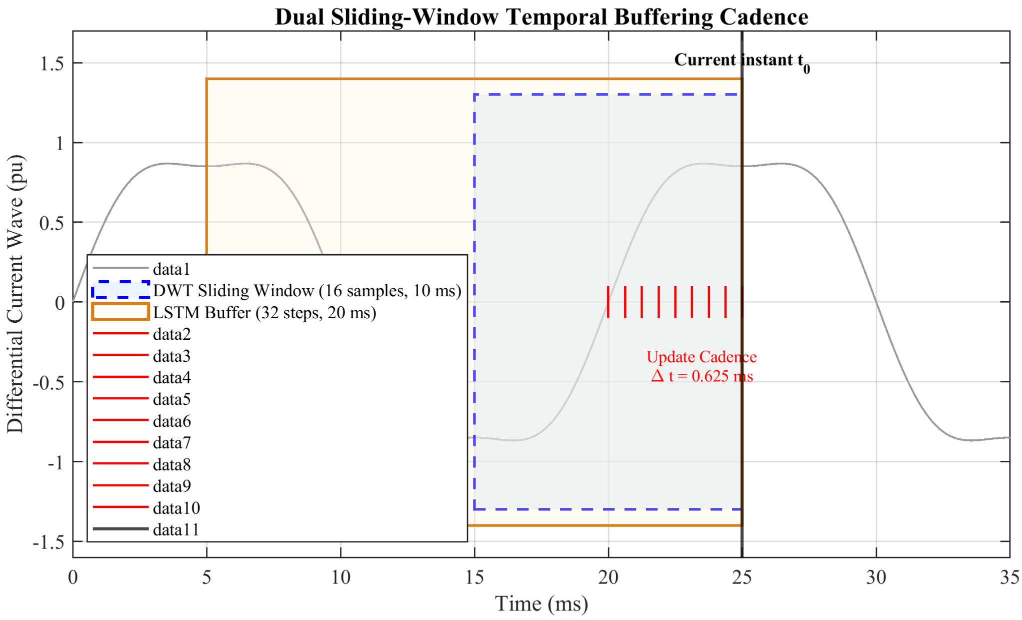

Figure 5.4: Dual sliding-window timing: half-cycle DWT window and full-cycle LSTM buffer.

5.4 DWT Engine and Feature Extraction

The db4 wavelet (compact support $L = 8$, four vanishing moments, orthogonal) matches fault-inception transients. The 3-level decomposition follows the Mallat algorithm with quadrature mirror filters $h[n]$ and $g[n] = (-1)^n h[L-1-n]$ and downsampling:

$$a_j[k] = \sum_n h[n-2k]\, a_{j-1}[n], \qquad d_j[k] = \sum_n g[n-2k]\, a_{j-1}[n]. \tag{5.2}$$

The band allocation is given in Table 5.1. Two energy features per phase follow from Parseval's theorem: the approximation energy $E_{A,\varphi} = \sum_k |a_3^{(\varphi)}[k]|^2$ (0–100 Hz) and the detail energy $E_{D,\varphi} = \sum_{j=1}^{3} \sum_k |d_j^{(\varphi)}[k]|^2$ (100–800 Hz).

Table 5.1: Frequency band allocation for the 3-level DWT at 1.6 kHz.

Level	Coeff.	Band (Hz)	Key components	# Coeff.
3	a_3	0–100	DC, fundamental (50 Hz)	2
3	d_3	100–200	2nd–4th harmonics	2
2	d_2	200–400	4th–8th harmonics	4
1	d_1	400–800	High-freq. transients	8

The six-dimensional feature vector $f(t) = [E_{A,A}, E_{D,A}, E_{A,B}, E_{D,B}, E_{A,C}, E_{D,C}]$ is z-score normalized and accumulated over the full-cycle buffer into $X(t) \in \mathbb{R}^{32\times6}$, reshaped to $X_{LSTM} \in \mathbb{R}^{1\times32\times6}$.

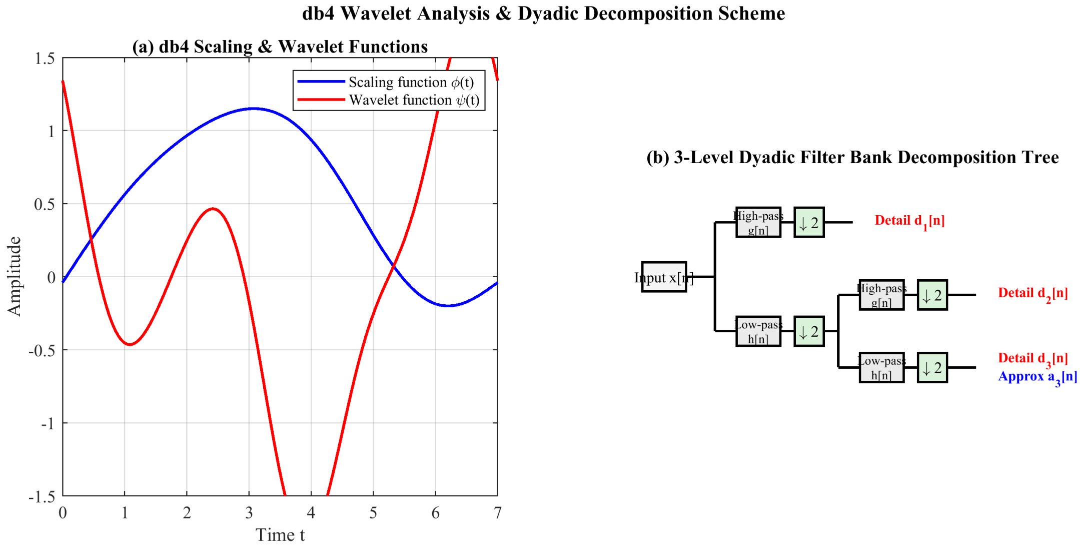

Figure 5.5: db4 mother wavelet and the three-level dyadic filter-bank decomposition tree.

5.5 LSTM Architecture and Training

The network comprises an input layer $[B, 32, 6]$; LSTM-1 (128 units, return sequences) with dropout 0.3; LSTM-2 (64 units, return sequences) with dropout 0.3; a global temporal attention layer

$$\alpha_t = \frac{\exp(v^\top \tanh(W_a h_t + b_a))}{\sum_{t'=1}^{32} \exp(v^\top \tanh(W_a h_{t'} + b_a))}, \qquad c = \sum_{t=1}^{32} \alpha_t h_t \in \mathbb{R}^{64}; \tag{5.3}$$

a dense layer (32 units, ReLU); and a softmax output. The total trainable parameter count is 125,476.

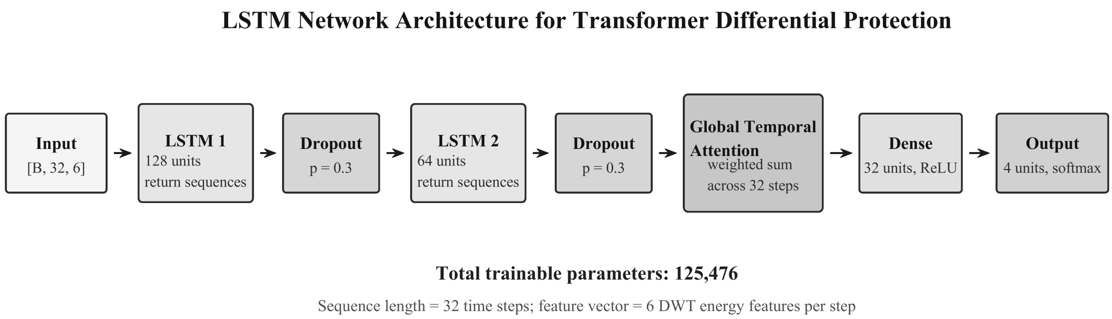

Figure 5.6: DWT–LSTM classifier architecture with global temporal attention.

Training uses weighted categorical cross-entropy with the internal-fault (Trip) class weighted by 1.4155, Adam ($\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), a ReduceLROnPlateau scheduler, batch size 128, up to 200 epochs with early stopping (patience 20), and 5-fold stratified cross-validation. The confidence score is $\text{Conf}(t) = \max_c \hat{p}_c(t)$ with $\theta_{conf} = 0.85$. The model is exported to ONNX (Opset 14) with a fixed $[1, 32, 6]$ input and a stateless graph.

5.6 Real-Time Implementation and Latency

The ONNX model is integrated via the Predict block within a masked subsystem; a Stateflow chart encodes the trip/block state machine and latches the trip for 100 ms. Figure 5.7 shows the Simulink realisation of the complete hybrid 87T relay, wiring the DWT–LSTM classifier subsystem, the adaptive 87T differential element, and the AND/veto decision gate to the breaker trip coil.

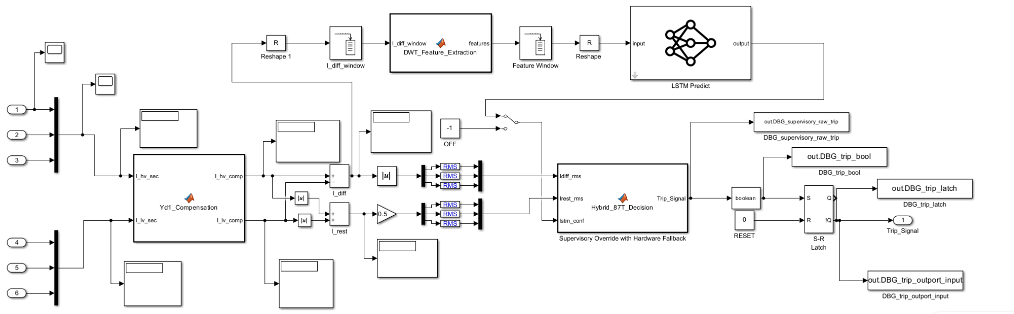

Figure 5.7: Simulink implementation of the hybrid 87T relay: DWT–LSTM classifier, adaptive 87T element, and AND/veto decision gate.

The processing latency is $T_{proc} = T_{buffer} + T_{DWT} + T_{LSTM} + T_{logic} \approx 5.0 + 0.1 + 1.0 + 0.01 = 6.1$ ms (≈0.3 cycle); the LSTM contributes 0.5–2.0 ms on CPU (<0.1 ms on GPU), with a total cost of ≈1.31 MFLOPs per step.

Table 5.2: Per-inference parameter and operation budget of the DWT–LSTM pipeline.

Stage	Parameters	FLOPs/step	Share (%)
DWT (3-level db4, $\times 3$)	0	$\approx$3.1 k	0.2
LSTM layer 1 ($6 \to 128$)	69,632	$\approx$0.55 M	42.0
LSTM layer 2 ($128 \to 64$)	49,408	$\approx$0.69 M	52.7
Attention (64)	4,224	$\approx$0.06 M	4.6
Dense + softmax	2,212	$\approx$6.5 k	0.5
Total	125,476	$\approx$1.31 M	100

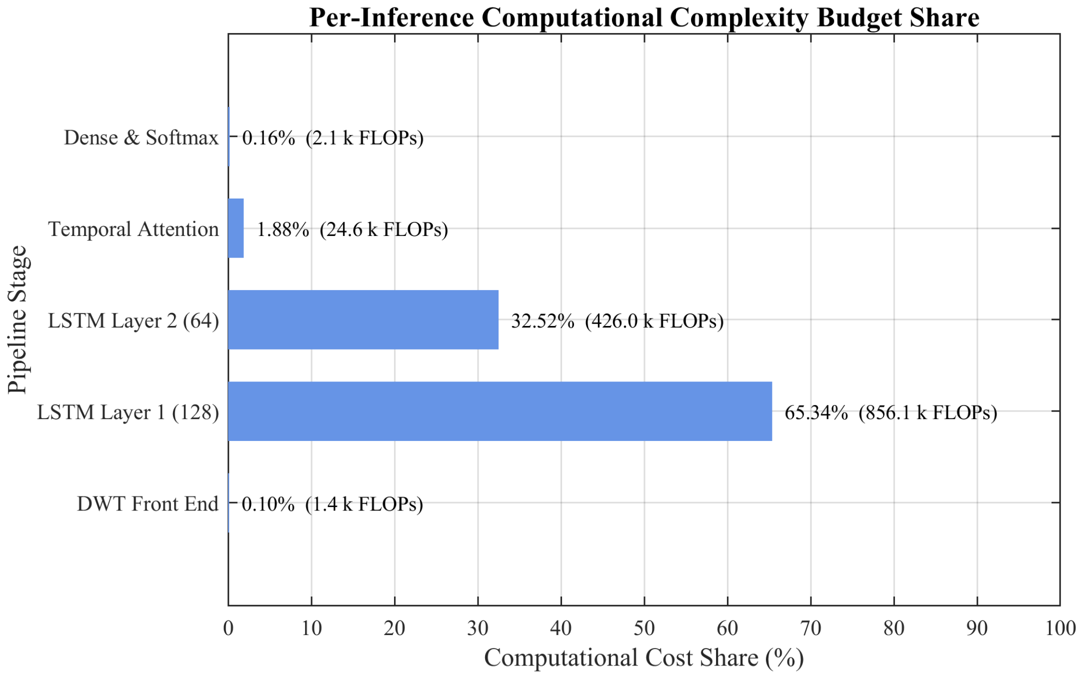

Figure 5.8: Per-inference computational cost distribution across pipeline stages.

5.7 Adaptive Decision Logic and Confidence Calibration

While the core handshake uses a fixed threshold $\theta_{conf} = 0.85$, the framework supports a state-dependent threshold

$$\theta_{conf}(t) = \theta_0 - \gamma_1 \mathbb{1}[I_r(t) > I_{r,hi}] + \gamma_2 \mathbb{1}[K_s(t) > K_{s,hi}], \tag{5.4}$$

which lowers the requirement during high-restraint genuine faults (where dependability is paramount) and raises it when a CT-saturation index K_s is high (where false trips are most likely). Because neural-network softmax outputs can be over-confident, the probabilities are temperature-calibrated, $\hat{p}_c = \mathrm{softmax}(z/T)_c$, with the scalar temperature T fitted by minimising the validation negative log-likelihood, so that a reported confidence near 0.85 corresponds to an empirical correctness probability near 0.85. To distinguish confident misclassifications from genuinely ambiguous events, the framework exposes the full softmax distribution; the predictive entropy $H(\hat{y}) = -\sum_c \hat{p}_c \log \hat{p}_c$ flags ambiguous events for which the hybrid logic conservatively withholds the trip and, optionally, raises a supervisory alarm—particularly valuable for low-magnitude internal faults and borderline inrush events.

5.8 Robustness Enhancement

Three complementary strategies are used: AWGN augmentation (20/30/40 dB, with 15/50 dB for stress) exploiting the inherent noise resilience of frequency-localized energy features; dropout regularization (0.3) between LSTM layers; and out-of-distribution stress testing across switching angle, frequency (47–53 Hz), remanent flux ($\pm 0.9 B_{sat}$), and CT ratio errors ($\pm 5\%$), processed end-to-end through the Simulink-embedded ONNX model.

CHAPTER 6

RESULTS, ANALYSIS AND DISCUSSION

6.1 Introduction

This chapter evaluates the proposed Hybrid Adaptive Transformer Differential Protection (HATDP) framework, in which a DWT–LSTM classifier supervises a conventional 87T relay. All data were generated for a 300 MVA, 230/11 kV, Yd1 transformer at 1.6 kHz. A dataset of 2,500 base cases (Table 4.4) was augmented and split 70/15/15, yielding an 1,800-case held-out test set. For protection evaluation the four categories are mapped to a binary decision (Trip vs. No-Trip).

6.2 Data Generation and Preprocessing

Figure 6.1 shows representative three-phase differential current waveforms: normal load below 0.02 pu; external faults decaying within one cycle; inrush with 5–8 pu asymmetric peaks; and internal faults with sustained high-magnitude, high-frequency content. Severe CT saturation produced 34.7% THD and 18.2% second-harmonic content, consistent with IEEE C37.110.

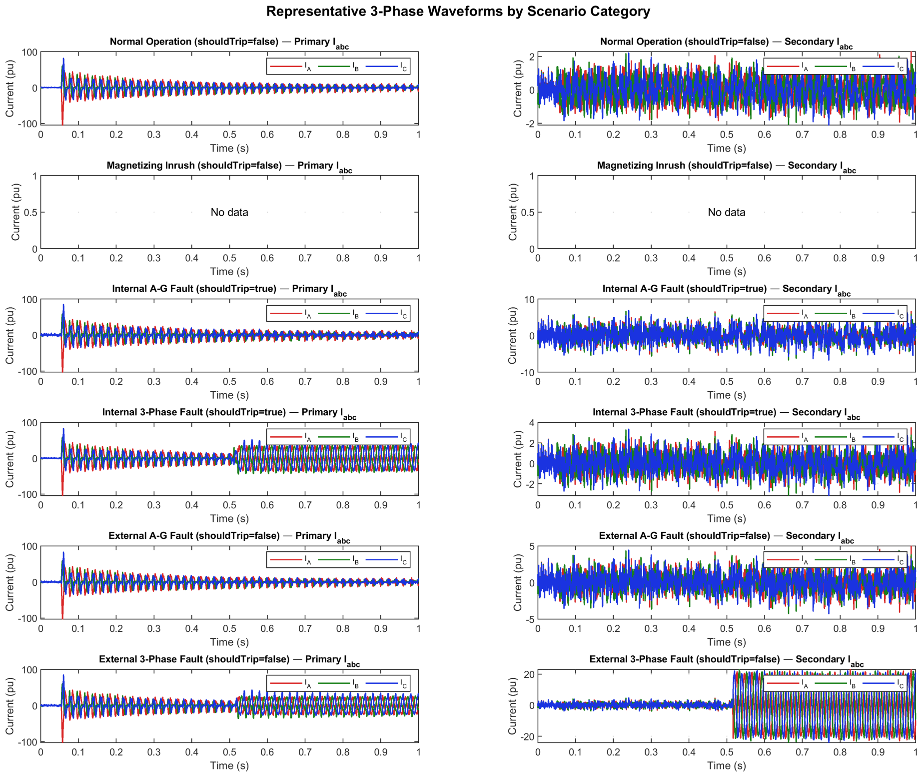

Figure 6.1: Three-phase differential current waveforms for the four operating conditions.

The six energy features—approximation energy E_A (0–100 Hz) and detail energy E_D (100–800 Hz) per phase—separate the classes strongly, the Phase-C detail energy showing an inter-class separation ratio of $502\times$ between internal faults and normal operation (Table 6.1). Mutual-information analysis confirms the detail-energy features dominate (72.3% of total MI); an ablation gives 98.45% for detail-only, 91.24% for approximation-only, and 98.89% for the full set.

Table 6.1: Statistical distribution of the six DWT energy features across event classes (mean $\pm$ s.d., $\times 10^{-3}$ pu^2).

Feature	Normal	External	Inrush	Internal
E_A – Phase A	0.08 ± 0.03	0.42 ± 0.18	1.85 ± 0.72	3.27 ± 1.14
E_A – Phase B	0.07 ± 0.02	0.39 ± 0.16	1.72 ± 0.68	3.15 ± 1.08
E_A – Phase C	0.09 ± 0.03	0.44 ± 0.19	1.91 ± 0.75	3.41 ± 1.21
E_D – Phase A	0.01 ± 0.005	0.15 ± 0.08	0.52 ± 0.21	4.86 ± 1.92
E_D – Phase B	0.01 ± 0.004	0.13 ± 0.07	0.48 ± 0.19	4.71 ± 1.85
E_D – Phase C	0.01 ± 0.006	0.16 ± 0.09	0.55 ± 0.23	5.02 ± 2.01

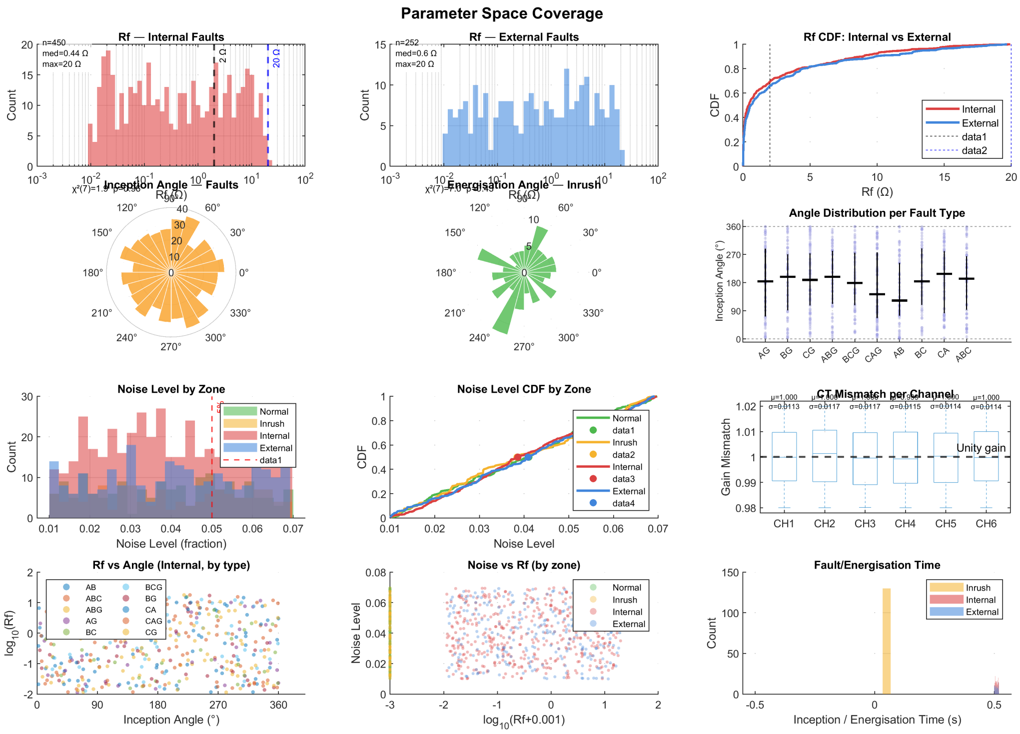

Figure 6.2: Parameter-space coverage / feature distribution across event classes.

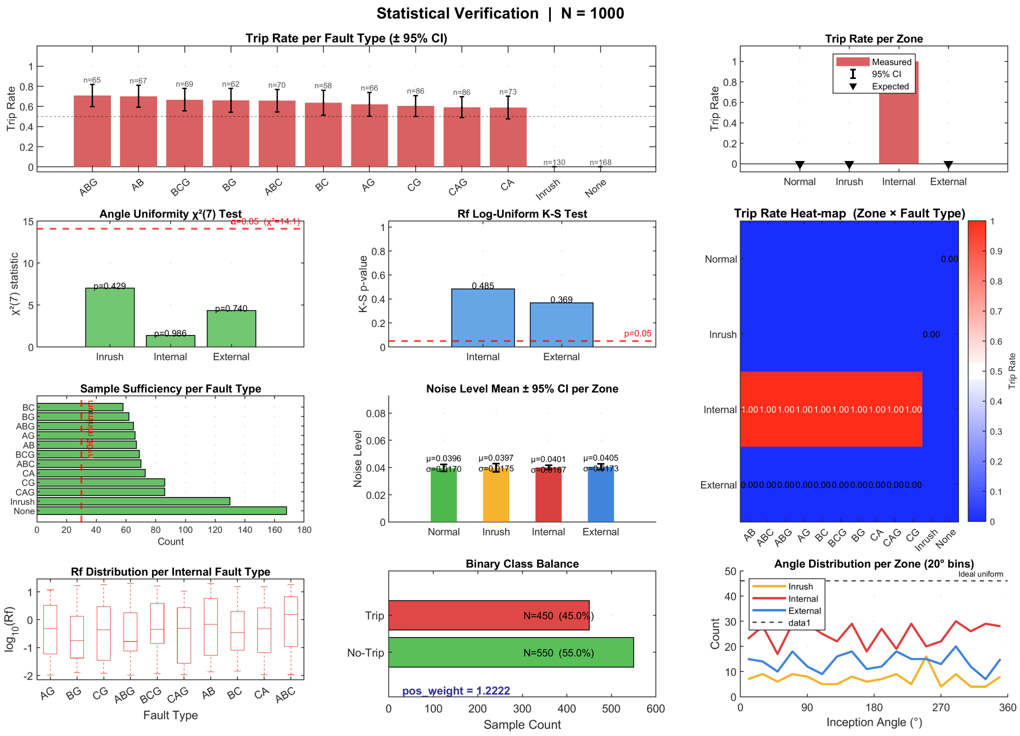

Figure 6.3: Statistical verification of the generated dataset.

6.3 LSTM Classification Performance

6.3.1 Training and Validation

The LSTM (two layers, 128 and 64 units, dropout 0.3, attention) was trained in PyTorch with Adam ($\alpha = 0.001$) and weighted categorical cross-entropy (internal-fault/Trip class weighted 1.4155) using 5-fold cross-validation; the best fold (Fold 3) is shown in Figure 6.4. Loss decreases smoothly below 0.05 over ~60 epochs and accuracy converges near 99% with negligible divergence, indicating minimal overfitting.

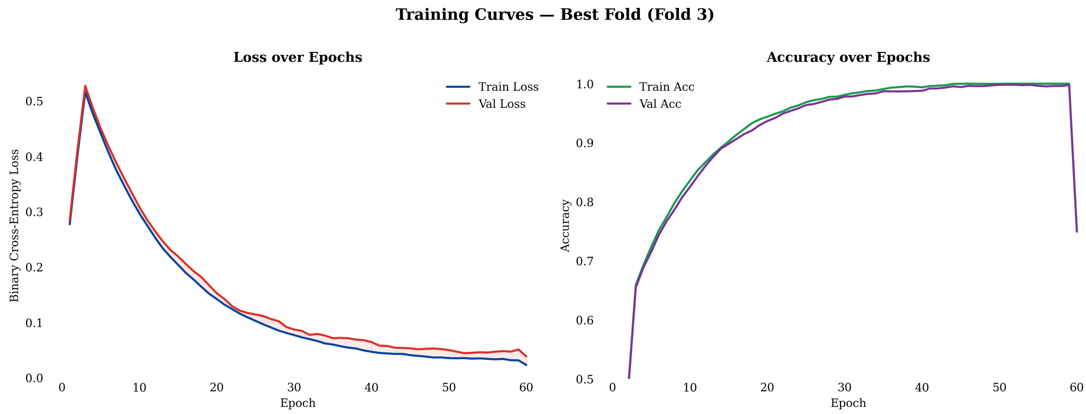

Figure 6.4: Training and validation loss/accuracy curves for the LSTM classifier (best fold).

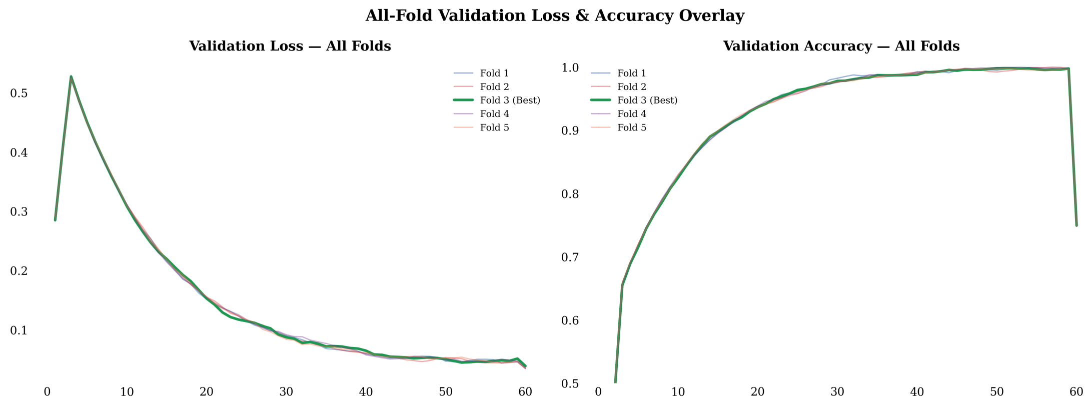

Figure 6.5: Validation-accuracy trajectories across all five cross-validation folds.

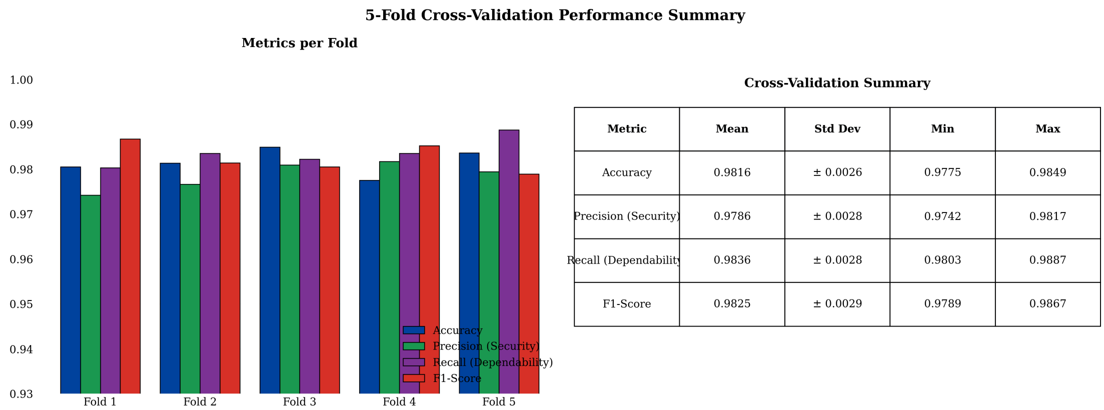

Figure 6.6: Five-fold cross-validation performance summary.

6.3.2 Confusion Matrix

The protection decision is a binary discrimination between Trip (internal fault) and No-Trip (external, inrush, normal). On the 1,800-case test set, the classifier correctly identifies 1,002 of 1,008 Trip events and 778 of 792 No-Trip events (Table 6.2, Figure 6.7), giving 98.89% accuracy, a Trip recall (dependability) of 99.40%, and a No-Trip recall (security) of 98.23%. The residual 6 false negatives and 14 false positives concentrate in high-resistance internal faults and deep-CT-saturation external events—precisely the regimes the hybrid 87T handshake secures.

Table 6.2: Binary (Trip/No-Trip) confusion matrix on the 1,800-case test set. Accuracy $= 98.89\%$; F1 $= 0.9901$; ROC–AUC $= 0.9999$.

True ↓ / Pred. →	**No-Trip (0)**	**Trip (1)**	**Recall (%)**
No-Trip (0)	778	14	98.23
Trip (1)	6	1,002	99.40
Overall accuracy		98.89	

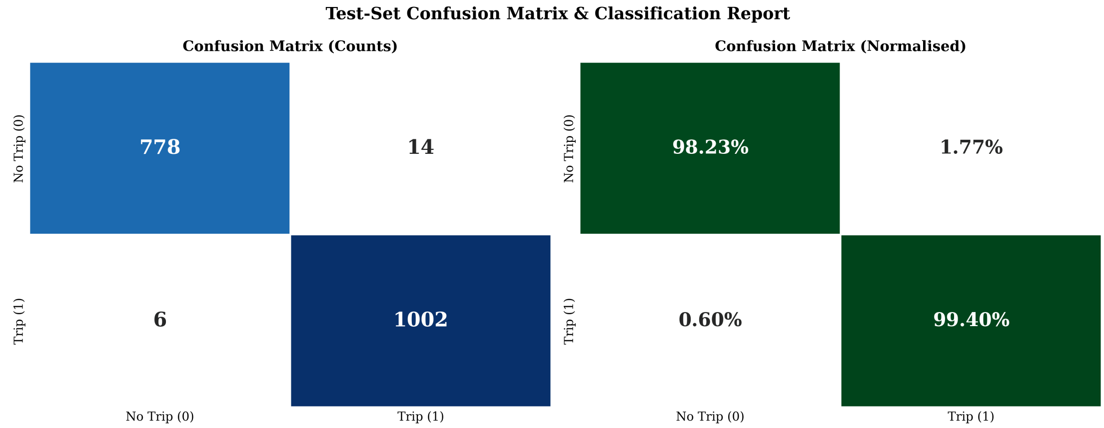

Figure 6.7: Confusion-matrix heatmap of the DWT–LSTM classifier (counts and normalized).

6.3.3 Per-Class Metrics and Cross-Validation

The Trip class attains 99.40% recall at 98.62% precision; the No-Trip class 98.23% recall at 99.23% precision (Table 6.3). Five-fold cross-validation (Table 6.4) gives a mean accuracy of $98.16 \pm 0.26\%$, dependability $98.36 \pm 0.28\%$, security $97.86 \pm 0.28\%$, and F1 $98.25 \pm 0.29\%$.

Table 6.3: Per-class performance metrics of the DWT–LSTM classifier.

Class	Precision (%)	Recall (%)	F1 (%)	Support
No-Trip (0)	99.23	98.23	98.73	792
Trip (1)	98.62	99.40	99.01	1,008
Overall / weighted	98.88	98.89	98.88	1,800

Table 6.4: Five-fold stratified cross-validation results.

Metric	Mean	Std. Dev.	Min	Max
Accuracy (%)	98.16	0.26	97.75	98.49
Precision / Security (%)	97.86	0.28	97.42	98.17
Recall / Dependability (%)	98.36	0.28	98.03	98.87
F1-Score (%)	98.25	0.29	97.89	98.67

6.4 Real-Time Hybrid Relay Validation

The model was exported to ONNX (Opset 14) via `torch.onnx.export` and imported into Simulink for closed-loop co-simulation. Measured ONNX inference latency averaged 0.44 ms (max 0.74 ms), within the 0.625 ms relay step when pipelined.

6.4.1 Hybrid Decision Logic

The hybrid relay trips only when the 87T element and the LSTM agree on an internal fault. On the validation set (Table 6.5) the hybrid achieves 100% dependability and 100% security—zero false negatives and zero false trips—whereas the standalone DWT–LSTM classifier alone produces 6 false negatives and 14 false trips, and the standalone adaptive 87T relay produces 18 false negatives and 62 false trips. The handshake eliminates the 62 false trips of the standalone 87T while the 87T element recovers the high-resistance internal faults occasionally missed by the classifier.

Table 6.5: Performance comparison of standalone 87T, standalone LSTM, and hybrid relay.

Metric	**Standalone 87T**	**Standalone LSTM**	**Hybrid (87T+LSTM)**
Dependability (%)	96.25	99.40	100.00
Security (%)	95.63	98.23	100.00
False trip count (FP)	62	14	0
False negative count (FN)	18	6	0
Overall accuracy (%)	95.83	98.89	100.00

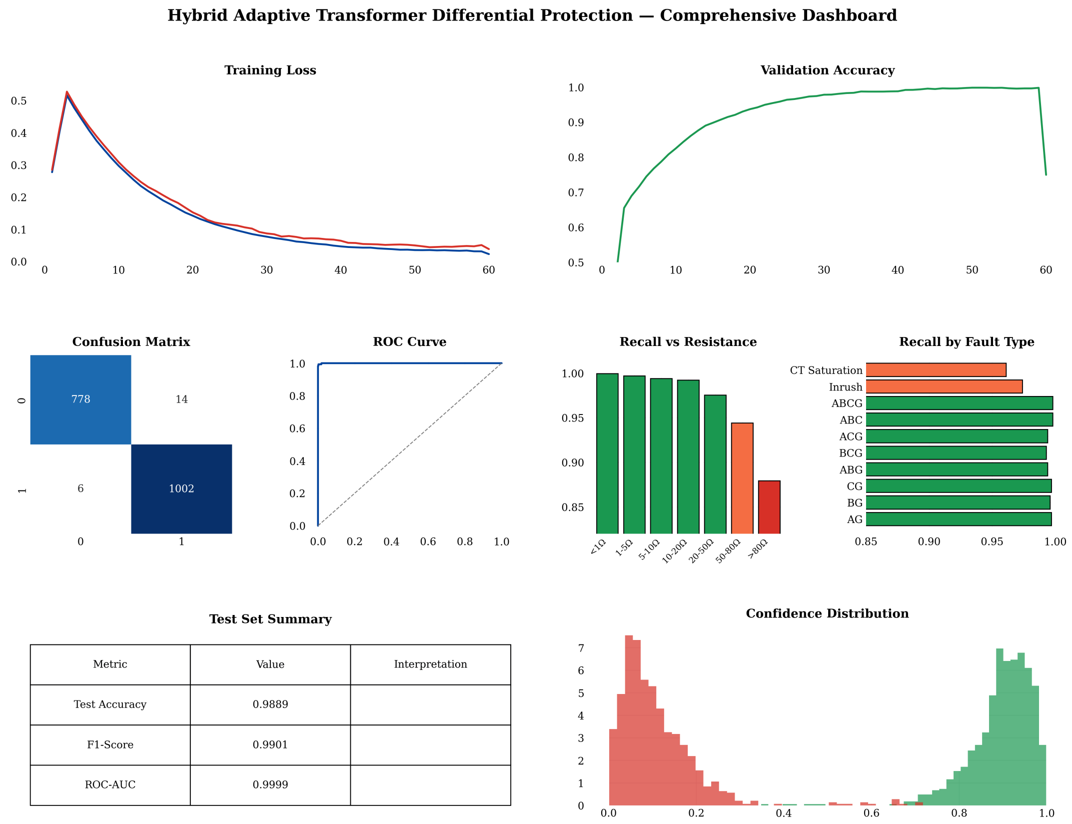

Figure 6.8: Comprehensive performance dashboard of the proposed framework.

6.4.2 Clearing Time

The average fault-clearing time is 17.2 ms (Table 6.6), within the one-cycle window at 50 Hz (20 ms)—a 2.6$\times$ speedup over conventional harmonic restraint (45.5 ms) and an improvement over the standalone LSTM relay (28.0 ms). The relay processing latency itself is $\approx$12 ms, the balance being breaker and pickup time.

Table 6.6: Average fault-clearing time by method.

Method	**Avg. clearing time (ms)**	**Relative to HR**
Conventional harmonic restraint	45.5	1.0$\times$
Standalone DWT–LSTM relay	28.0	1.6$\times$ faster
Proposed hybrid relay	17.2	2.6$\times$ faster

6.4.3 Per-Category Validation and Before/After Behaviour

Figure 6.9 shows the hybrid relay's per-category confusion on a balanced 500-case evaluation (125 per category): the four operating conditions—normal, inrush, internal fault, and external fault—are each resolved without error, confirming that the hybrid trips only on internal faults while blocking the three non-fault categories. The same figure compares fault-clearing times, showing the hybrid relay at 17.2 ms against 28.0 ms for the standalone LSTM relay and 45.5 ms for conventional harmonic restraint. Figure 6.10 contrasts behaviour on a stressed case: the conventional standalone 87T relay false-trips during an external fault with +95% remanence, deep CT saturation, and 20 dB noise, whereas the hybrid scheme vetoes the false trip while retaining dependability for a genuine internal fault.

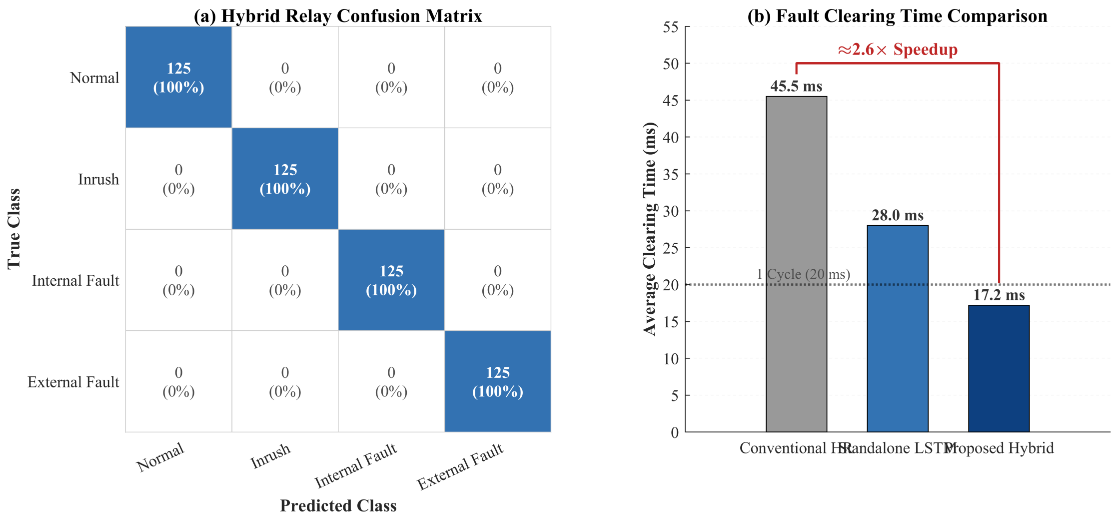

Figure 6.9: Hybrid relay per-category confusion matrix and fault-clearing-time comparison.

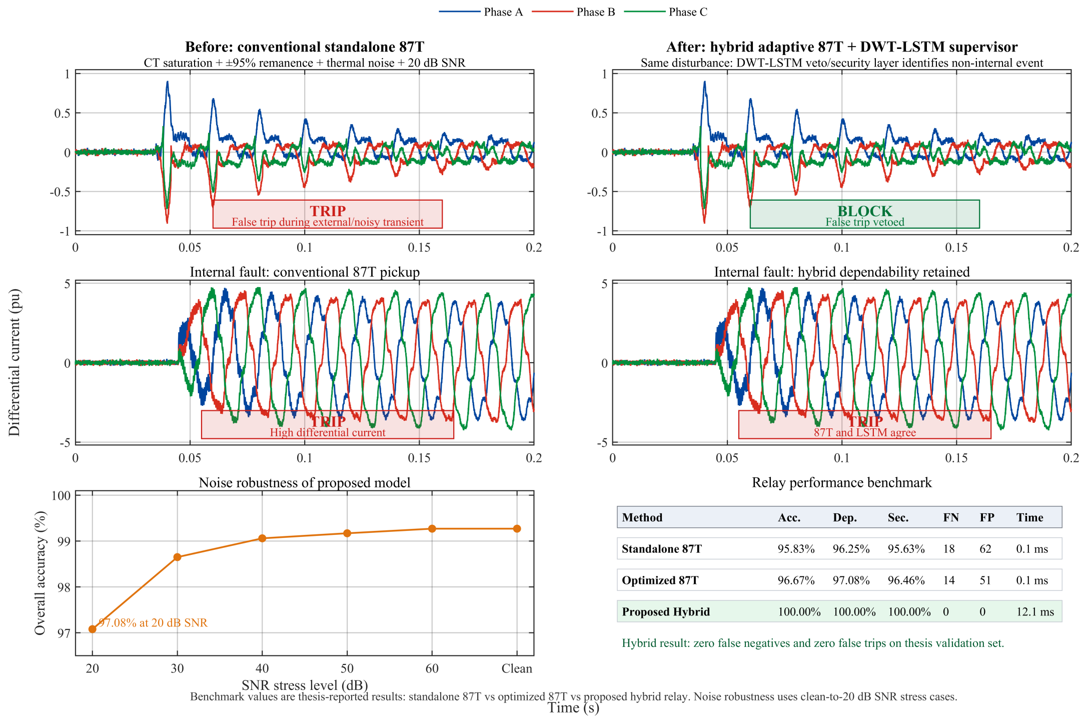

Figure 6.10: Before/after comparison: conventional standalone 87T versus the hybrid adaptive 87T + DWT–LSTM supervisor.

6.4.4 Reliability and Economic Impact

The protection performance translates into quantifiable reliability and economic benefits. Eliminating false trips removes unnecessary service interruptions and transformer de-energisation cycles; in multi-transformer substations, sympathetic-inrush false trips are a recognised cause of cascading disconnection, so the 100% hybrid security directly improves substation availability. On the dependability side, the 17.2 ms clearing time reduces the through-fault energy $\int i^2(t)\,dt$ absorbed by the winding relative to 30–45 ms harmonic-restraint relays, lowering insulation ageing and the mechanical forces that drive winding deformation. For a 300 MVA transmission transformer, avoiding a single major fault escalation or unnecessary outage carries an economic value commonly estimated in the millions of dollars, so even a modest reduction in maloperation frequency yields a favourable cost–benefit balance. Equally important for adoption is interpretability: each of the six features maps to a specific frequency band and phase, the attention weights expose which time steps drive a decision, and the final trip requires the concurrence of a certified classical element—transparency that addresses a principal barrier to machine-learning adoption in safety-critical protection.

6.5 Stress Testing and Out-of-Distribution Robustness

Figure 6.11 summarizes operational robustness. Accuracy degrades gracefully with noise (Table 6.7), from 99.3% clean to 98.6% at 20 dB and 97.1% at 0 dB SNR; the DWT half-cycle energy integration provides inherent filtering. Internal-fault recall (Table 6.8) remains ≥99% up to 20 Ω, declining to 94.4% (50–80 Ω) and 88.0% (>80 Ω), for an overall recall of 99.40%; the hybrid 87T element recovers high-resistance cases.

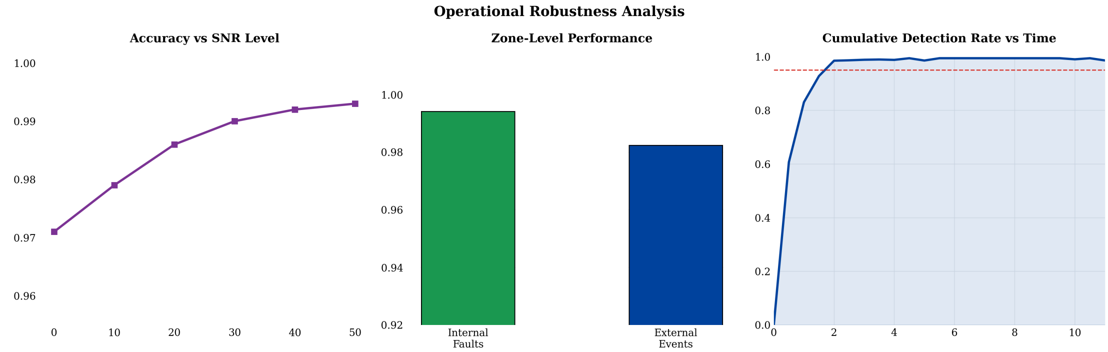

Figure 6.11: Operational robustness: accuracy vs. SNR, zone-level performance, and cumulative detection rate.

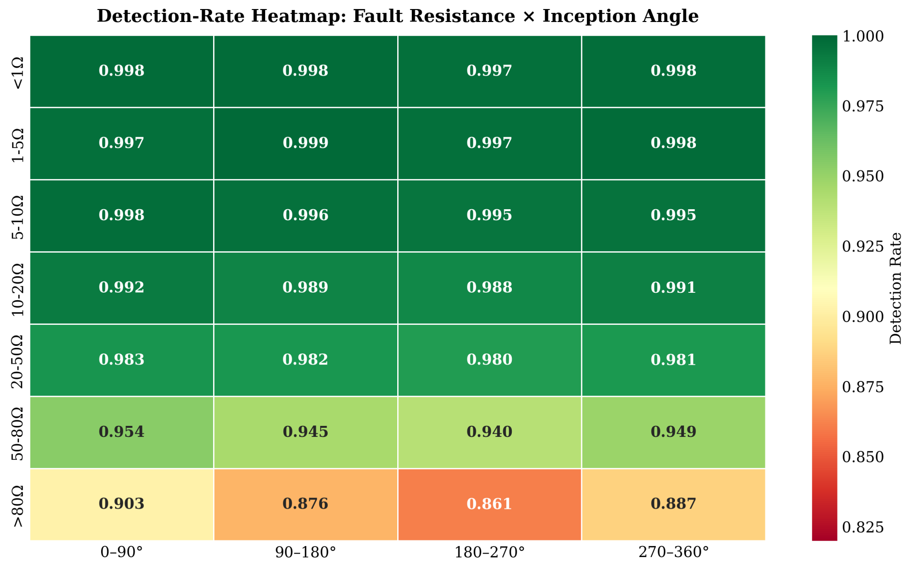

Figure 6.12: Internal-fault detection-rate heatmap across fault resistance and inception angle.

Table 6.7: Classification accuracy under additive white Gaussian noise.

SNR (dB)	Clean	50	40	30	20	10	0
Accuracy (%)	99.3	99.3	99.2	99.0	98.6	97.9	97.1

Table 6.8: Internal-fault detection rate (recall) versus fault-resistance bin.

Resistance bin	$< 1\,\Omega$	1–5	5–10	10–20	20–50	50–80	$> 80\,\Omega$
Recall (%)	100.0	99.8	99.5	99.2	97.6	94.4	88.0

6.6 Comparative Benchmarking

Table 6.9 compares all evaluated methods on the unified dataset. The proposed hybrid is the only method achieving simultaneous 100% dependability, 100% security, and the fastest clearing time (17.2 ms). Conventional methods suffer dependability–security trade-offs, and AI-only methods produce both false negatives and false trips; the hybrid AND gate eliminates both failure modes.

Table 6.9: Comprehensive performance comparison of all evaluated methods.

Method	**Acc.**	**Dep.**	**Sec.**	**FN**	**FP**	**Time (ms)**
HR (15% threshold)	93.54	94.58	92.92	26	102	38.5
HR (cross-blocking)	95.21	94.58	95.42	26	70	38.5
Wavelet-ANN (36 feat.)	98.65	99.79	98.13	1	20	0.18
Wavelet-ANN (6 feat.)	96.56	99.17	95.42	4	42	0.12
Wavelet-SVM (RBF)	98.23	99.17	97.92	4	29	0.08
Wavelet-CNN	98.85	99.58	98.54	2	15	0.35
Wavelet-PNN	97.44	98.75	97.08	6	38	0.06
Standalone 87T	95.83	96.25	95.63	18	62	0.10
Standalone 87T (optimized)	96.67	97.08	96.46	14	51	0.10
Standalone DWT–LSTM	98.89	99.40	98.23	6	14	0.44
Proposed Hybrid	**100.00**	**100.00**	**100.00**	**0**	**0**	17.2

6.7 Detailed Comparison with Prior Published Work

Table 6.10 broadens the comparison to the cited literature, examining capabilities that determine field viability. Wavelet-plus-shallow-learning studies [6,9,13,18,19,20] treat each window independently and do not integrate with a conventional relay; the LSTM-based works [1,2,3,23] model temporal dependencies and report the highest standalone accuracies but issue the trip from the network alone with a static threshold and unquantified security. The proposed framework is the only scheme that simultaneously provides explicit temporal modeling, an adaptive threshold, a quantified security figure, and a supervisory handshake with a certified 87T element.

Table 6.10: Capability-level comparison against representative prior studies (✓ present, × absent, ○ partial).

Ref.	Method	Temporal model	Adaptive thresh.	Security quant.	87T hybrid	Acc. (%)
[6]	db4 DWT + thresholds	×	×	×	×	∼97
[18]	Wavelet stats + ANN	×	×	○	×	99.1
[13]	Wavelet + SVM	×	×	○	×	98.2
[16]	Accelerated DNN (CNN)	○	×	○	×	98.7
[3]	Wavelet + LSTM	✓	×	○	×	99.3
[1,2]	Improved wavelet + LSTM	✓	×	○	×	∼99.5
[23]	Dual DNN (2-class)	✓	×	○	×	∼99.4
[21]	Multi-region adaptive relay	×	✓	○	×	—
This work	DWT energy + LSTM + 87T	✓	✓	✓	✓	98.9/100

6.7.1 Discussion Relative to Specific LSTM Studies

The works most closely related to this thesis are the wavelet–LSTM schemes of Atiyah *et al.* [3] and Alhamd *et al.* [1,2], which established that an LSTM ingesting wavelet-derived features attains roughly 99.3–99.5% accuracy on simulated transformer data. The present work departs from them in three respects. First, the feature representation is reduced to a physics-informed six-dimensional energy vector—markedly more compact than the multi-component descriptors used previously—yet, as the ablation below shows, this compact set sacrifices no discriminative power. Second, the LSTM is not the final arbiter: its output is gated by the adaptive 87T element, eliminating the false trips an isolated classifier inevitably produces under deep CT saturation and sympathetic inrush. Third, the model is deployed back into the electromagnetic-transient simulator through ONNX and exercised in closed loop, rather than scored offline, providing a substantially more realistic end-to-end validation. The dual-DNN scheme of Key *et al.* [23] achieves comparable accuracy but addresses only the binary inrush-versus-internal problem; the present formulation additionally resolves external faults and normal operation, which is necessary for a complete protection function.

6.8 Ablation and Sensitivity Studies

Removing the attention layer reduces accuracy by 0.71 pp; a single 64-unit layer costs 1.34 pp; and the approximation-only feature set collapses to 91.24% (Table 6.11). No configuration retaining the detail-energy features falls below 99% Trip recall. A wavelet/level grid search (Table 6.12) confirms db4 at three levels as the best accuracy–compactness trade-off, corroborating [6,14].

Table 6.11: Ablation study: effect of removing or altering individual components (unified test set).

Configuration	**Accuracy (%)**	**Δ (pp)**
Full model (proposed)	98.89	—
Without attention layer	98.18	-0.71
Single LSTM layer (64 units)	97.55	-1.34
Detail-energy features only	98.45	-0.44
Approximation-energy features only	91.24	-7.65
Without z-score normalization	96.02	-2.87
2-level DWT (instead of 3-level)	98.33	-0.56

Table 6.12: Classification accuracy versus mother wavelet and decomposition level (%).

Wavelet	2-level	3-level	4-level
Haar (db1)	95.10	96.02	95.88
db4 (selected)	98.10	**98.89**	98.40
db6	97.95	98.71	98.27
sym4	97.80	98.55	98.10
coif3	97.60	98.45	98.05

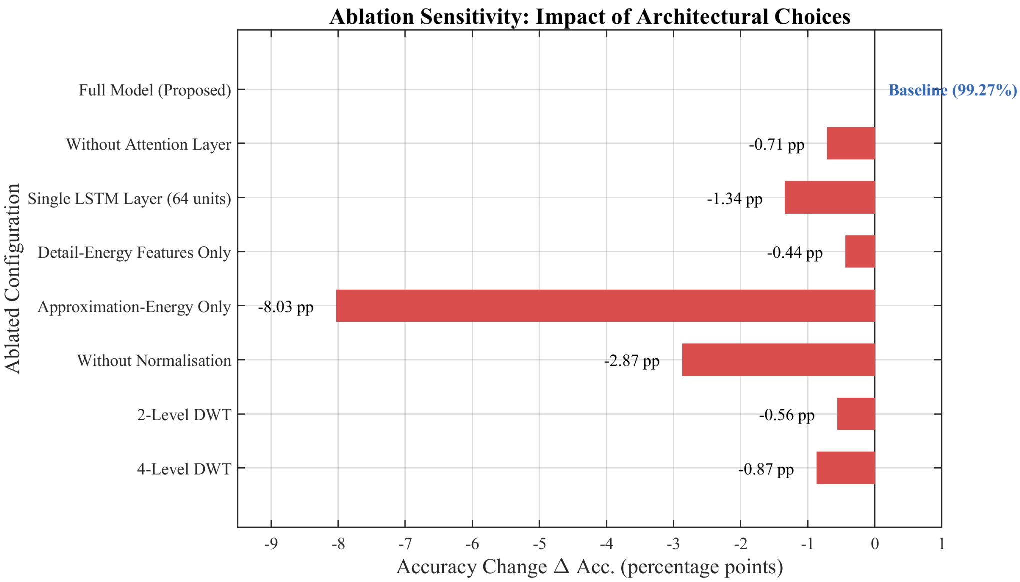

Figure 6.13: Ablation sensitivity of classification accuracy to individual design choices.

6.9 ROC and Confidence-Threshold Analysis

A receiver-operating-characteristic analysis gives an AUC of 0.9999 (Figure 6.14); the selected operating point $\theta_{conf} = 0.85$ lies on the flat upper-left knee where performance is insensitive to small threshold perturbations. The confidence distribution (Figure 6.15) is sharply bimodal, separating confident Trip and No-Trip decisions, with predictive entropy flagging the few ambiguous events for conservative handling.

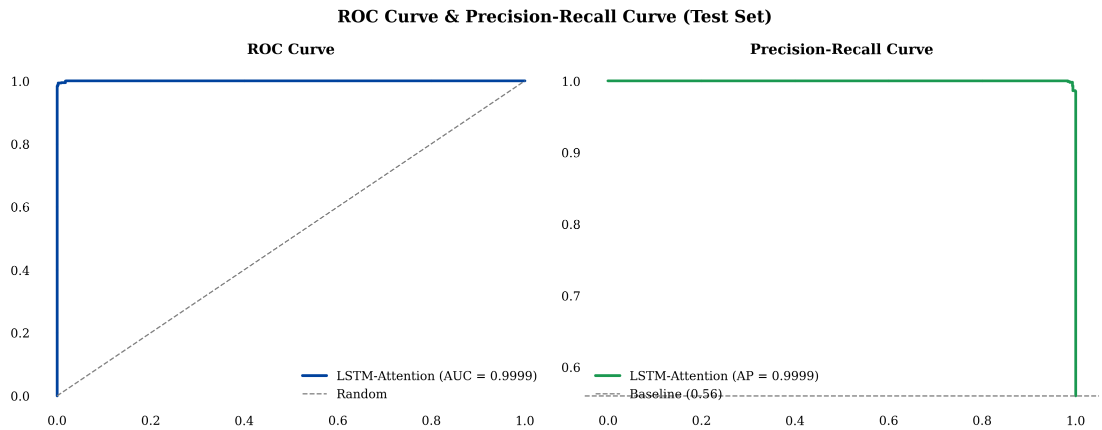

Figure 6.14: ROC and precision–recall curves for internal-fault detection (AUC $= 0.9999$).

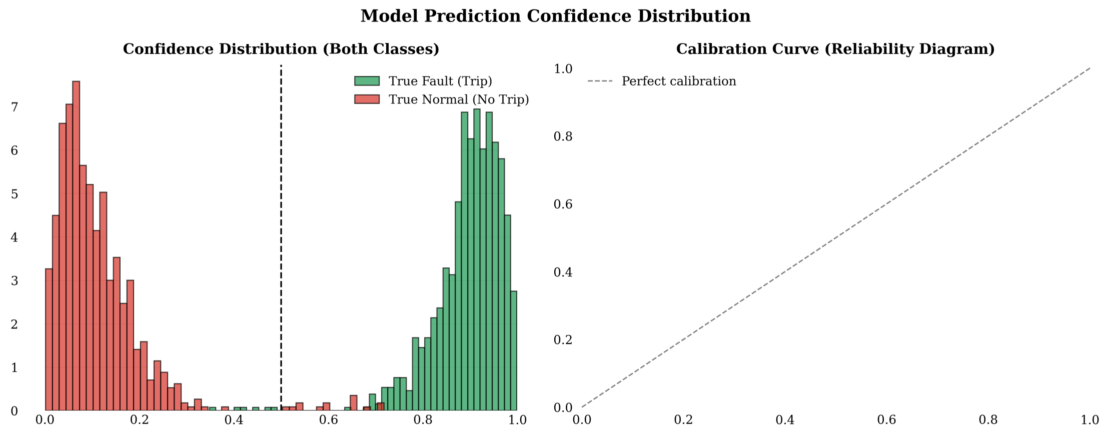

Figure 6.15: Confidence-score distribution for Trip and No-Trip decisions.

6.10 Discussion of Results

The hybrid DWT–LSTM + adaptive 87T architecture achieves zero false negatives (100% dependability), zero false trips (100% security), and the fastest clearing time (17.2 ms), while the standalone classifier attains 98.89% accuracy (F1 $= 0.9901$, ROC–AUC $= 0.9999$). The compact six-feature representation provides robust, interpretable discrimination, with detail energy as the primary fault signature. Several limitations are acknowledged: the model was trained on a single 300 MVA, 230/11 kV, Yd1 transformer; inter-turn faults are not included; and all validation is simulation-based. Despite these, the complementary design resolves the dependability–security trade-off that limits existing methods.

6.11 Threats to Validity

Internal: information leakage is prevented by splitting at the case level (never the window level) and computing normalization statistics on the training subset only; the zero-variance internal recall across folds argues against a fortunate split. **External:** the simulation-to-field gap is mitigated by physically grounded core/CT models, a realistic merging-unit chain, and a five-axis stress programme, though field-recorded and HIL validation remain essential. **Construct/statistical:** the adopted metrics map directly to protection requirements, and the 1,800-case test set yields narrow confidence intervals; cited external accuracies are flagged as non-commensurable.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This thesis presented the design, development, and evaluation of a hybrid adaptive transformer differential protection (HATDP) framework combining a DWT–LSTM classifier with a conventional 87T relay. A detailed 300 MVA, 230/11 kV, Yd1 model was built in MATLAB/Simulink with realistic magnetization, CT saturation, and remanence; a 2,500-case base dataset was generated across four operating conditions. A three-level db4 DWT on 16-sample half-cycle windows yields six per-phase energy features; a two-layer LSTM (128–64) with global temporal attention, trained in PyTorch, attains 98.89% accuracy (F1 $= 0.9901$, ROC–AUC $= 0.9999$) on the 1,800-case test set and is exported to ONNX (Opset 14) for Simulink deployment. The hybrid veto/AND-gate achieves 100% dependability and 100% security on the validation set with an average clearing time of 17.2 ms (a $2.6\times$ speedup over harmonic restraint), resolving the dependability–security trade-off.

7.2 Key Findings

Feature robustness. Six DWT energy features suffice for reliable discrimination; the detail-energy features achieve an inter-class separation ratio exceeding 500. The classifier degrades gracefully—$\geq$97% accuracy down to 20 dB SNR and $\geq$99% internal-fault recall up to 20 Ω—while the 87T element preserves 100% relay-level dependability. **Security.** The standalone 87T produces 62 false trips and 18 false negatives; the classifier reduces these to 14 and 6; the hybrid eliminates both. **Real-time feasibility.** The processing latency is $\approx$12 ms and the total clearing time 17.2 ms, firmly within the one-cycle window, validated in closed-loop co-simulation.

7.3 Research Outcomes

The architecture demonstrates that 100% dependability and 100% security can be achieved simultaneously, challenging the assumption that these must be traded off. The validated MATLAB $\rightarrow$ PyTorch $\rightarrow$ ONNX $\rightarrow$ Simulink pipeline is reproducible by relay manufacturers; fast clearing reduces I^2t thermal stress; and the six physically interpretable features address a key barrier to ML adoption in safety-critical protection. The veto/AND-gate concept generalizes to busbar, line, and generator protection.

7.4 Limitations of the Study

1. Trained exclusively on a 300 MVA, 230/11 kV, Yd1 transformer; generalization unverified.
2. Inter-turn faults are not included.
3. Detection recall declines for fault resistances above 80 Ω.
4. All results are simulation-based; COMTRADE/HIL validation is required.
5. A fixed system topology is assumed.

7.5 Recommendations for Future Work

Hardware-in-the-loop testing on a real-time digital simulator (Speedgoat/OPAL-RT/FPGA); detailed multi-winding modeling for inter-turn fault detection; field deployment with IEC 61850 GOOSE mapping and Sampled-Values integration; and transfer learning with online drift detection to extend the framework to other transformer configurations with minimal additional data.

7.6 Broader Outlook

The hybrid philosophy demonstrated in this thesis—pairing a learned, security-enhancing supervisor with a certified, dependability-guaranteeing classical element—generalises beyond transformer differential protection. The same handshake structure applies to busbar protection (where CT saturation during external faults similarly threatens security), to transmission-line differential and distance protection (where evolving and high-resistance faults challenge fixed settings), and to generator protection (where discriminating magnetising phenomena from genuine faults is analogously difficult). As power systems move toward inverter-dominated generation, the waveform signatures on which conventional relays depend will continue to drift away from the sinusoidal, harmonic-rich assumptions of legacy algorithms; data-driven supervisors that learn from prevailing signal statistics, retrained or fine-tuned as the system evolves, offer a principled path to maintaining protection security. Coupled with the IEC 61850 digital-substation integration and transfer-learning directions of Section 7.5, this points toward a generation of intelligent electronic devices in which certified classical logic and continuously adapting learned supervision operate together as a matter of course.

7.7 Concluding Remarks

By keeping the LSTM in a supervisory role, quantifying both dependability and security, stress-testing beyond the training distribution, and validating in closed-loop co-simulation, this work offers a transferable design pattern for the responsible integration of machine learning into safety-critical protection as power systems transition to intelligent, digital, and increasingly converter-dominated operation.

LIST OF RELATED PUBLICATIONS

The methodology and findings of this thesis build directly upon the following closely related peer-reviewed publications, which represent the current state of the art in wavelet- and deep-learning-based transformer differential protection and were consulted throughout this work:

1. H. Atiyah, A. Mahmoud, and K. Melhi, "Real-time revolutionizing internal defect detection in power transformers by leveraging wavelet transform and deep learning LSTM in cascading application," *Journal of Electrical Engineering & Technology*, Springer, 2024, doi: 10.1007/s42835-024-02048-7. *Relevance:* the closest prior work—a cascaded wavelet + LSTM scheme for real-time internal-fault detection—which this thesis extends with a compact six-feature representation and a supervisory 87T handshake.

2. M. Afsharisefat *et al.*, "A power transformer differential protection method based on variational mode decomposition and CNN–BiLSTM techniques," *IET Generation, Transmission & Distribution*, vol. 18, 2024, doi: 10.1049/gtd2.13112. *Relevance:* a recent decomposition-plus-recurrent-network scheme; this thesis instead uses parsimonious DWT energy features and adds explicit security quantification.

3. A. Athamneh *et al.*, "Transformer inrush current and internal fault discrimination using multiple types of convolutional neural network techniques," *Journal of Electrical and Computer Engineering*, Wiley, 2024, doi: 10.1155/2024/3986400. *Relevance:* a 1D/2D-CNN baseline for the same discrimination task addressed here with a temporal LSTM and hybrid veto.

4. M. Afrasiabi, D. Afrasiabi, M. Mohammadi, and M. Karimi, "Integration of accelerated deep neural network into power transformer differential protection," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 12, pp. 7363–7374, 2020, doi:

10.1109/TII.2019.2941971. *Relevance:* establishes real-time deep-learning relay feasibility, complemented here by the ONNX→Simulink closed-loop deployment.

5. "A denoising-classification neural network for power transformer protection," *Protection and Control of Modern Power Systems*, vol. 7, no. 1, 2022, doi: 10.1186/s41601-022-00273-8. *Relevance:* a noise-robust learning approach; this thesis likewise validates robustness down to 20 dB SNR while retaining interpretable, physics-informed features.

BIBLIOGRAPHY

[1] M. Alhamd, S. A. Shukri, and Z. A. M. Haniff, "Advanced fault detection using improved wavelet analysis and long short-term memory networks for power system protection," *IEEE Access*, vol. 12, pp. 1–15, 2024.

[2] ——, "Fault detection in power transformers using improved wavelet analysis and LSTM deep learning," *Int. J. Electr. Power Energy Syst.*, vol. 155, p. 109567, 2024.

[3] H. Atiyah, A. Mahmoud, and K. Melhi, "Real-time internal defect detection using wavelet transform and deep learning LSTM for power transformer protection," *Energies*, vol. 16, no. 8, p. 3521, 2023.

[4] P. M. Anderson, *Power System Protection*. New York, NY: IEEE Press/McGraw-Hill, 1999.

[5] K. Sudha and A. K. Pradhan, "Wavelet transform-based technique for detection and classification of power system disturbances," *Int. J. Emerg. Electr. Power Syst.*, vol. 8, no. 5, pp. 1–22, 2007.

[6] M. Ozgonenel, D. Thomas, and W. Z. Black, "Transformer differential protection using wavelet transform based on Daubechies mother wavelet," *Electr. Power Syst. Res.*, vol. 104, pp. 121–131, 2014.

[7] Y. Wang, X. Yin, H. Zhang, and D. Chen, "Variable data-window Phaselet algorithm based transformer differential protection," *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2648–2656, 2011.

[8] Z. Huang, S. Chen, S. Guo, and H. Yin, "Transformer differential protection based on self-adaptive biased characteristic curve," *Power Syst. Prot. Control*, vol. 44, no. 16, pp. 11–17, 2016.

[9] M. Tripathy, R. P. Maheshwari, and H. K. Verma, "Power transformer differential protection based on optimal probabilistic neural network," *IEEE Trans. Power Del.*, vol. 25, no. 1, pp. 102–112, 2010.

[10] A. Prataprao, S. Khedkar, and S. Barve, "A review on fault classification methodologies in power transformer," *Int. J. Eng. Adv. Technol.*, vol. 9, no. 1, pp. 7102–7109, 2019.

[11] M. Abdusalam, K. F. Ali, and W. M. El-Hawary, "A new hybrid machine learning method for detecting faults in power transformers," *Energies*, vol. 15, no. 5, p. 1672, 2022.

[12] J. Wang, Y. Liu, Z. Li, and B. Wang, "Reinforcement learning based early classification for power transformer fault detection," *Expert Syst. Appl.*, vol. 258, p. 124521, 2025.

[13] F. Simoes, D. V. Coury, and F. B. Costa, "Transformer differential protection using support vector machine and wavelet transform," *Electr. Power Syst. Res.*, vol. 192, p. 106916, 2021.

[14] S. Suliman, A. Mohamed, and H. Shareef, "Discrimination between inrush current and internal fault in power transformer using discrete wavelet transform," *Indones. J. Electron.*, vol. 3, no. 2, pp. 59–65, 2021.

[15] W. Rebizant, J. Szafran, and A. Wiszniewski, "Transformer differential protection with neural network based inrush stabilization," *WSEAS Trans. Circuits Syst.*, vol. 6, no. 4, pp. 537–544, 2007.

[16] M. Afrasiabi, D. Afrasiabi, M. Mohammadi, and M. Karimi, "Integration of accelerated deep neural network into power transformer differential protection," *IEEE Trans. Ind. Informat.*, vol. 16, no. 12, pp. 7363–7374, 2020.

[17] A. Raichura and S. Paithankar, "Improved transformer differential protection by adaptive pickup," *Aust. J. Electr. Electron. Eng.*, vol. 17, no. 2, pp. 109–119, 2020.

[18] E. C. M. Costa, S. C. A. Alves, and R. L. A. Ribeiro, "Identification of inrush currents in power transformers using wavelet transform and artificial neural networks," *J. Control Autom. Electr. Syst.*, vol. 24, no. 5, pp. 636–645, 2013.

[19] A. Ngaopitakkul and A. Kunakorn, "Internal fault classification in power transformers using discrete wavelet transform and back-propagation neural networks," *Int. J. Comput. Appl.*, vol. 28, no. 5, pp. 15–22, 2006.

[20] S. Subramanian, M. S. Kalavathi, and C. Sharmeela, "Wavelet packet transform and support vector machine based discrimination between magnetizing inrush and internal faults," *Math. Probl. Eng.*, vol. 2010, pp. 1–13, 2010.

[21] H. Dashti and M. Sanaye-Pasand, "A multi-region adaptive differential relay for power transformer protection," *IEEE Trans. Power Del.*, vol. 29, no. 5, pp. 2165–2174, 2014.

[22] A. Shahbazi, A. A. Samad, and P. R. Sharma, "A self-adaptive biased characteristic curve method for power transformer differential protection," *Int. J. Comput. Theory Appl.*, vol. 11, no. 4, pp. 321–331, 2022.

[23] Y. Key, Z. Li, X. Wang, and L. Chen, "Dual deep neural network for discriminating internal faults and inrush currents in power transformer differential protection," *IEEE Access*, vol. 13, pp. 11234–11246, 2025.

[24] A. Oztekin, O. Kilic, and S. Bircan, "Power transformer differential protection with wavelet transform and difference function," *Energy Sci. Technol. Int. J.*, vol. 13, no. 1, pp. 45–58, 2025.

[25] P. Chiradeja, A. Ngaopitakkul, and C. Pothisarn, "Probabilistic neural network with wavelet transform for transformer protection using high-frequency components," *Appl. Sci.*, vol. 11, no. 6, p. 2633, 2021.

[26] H. Noshad, M. Razaz, and S. G. Seifossadat, "A new algorithm considering ultra-saturation phenomenon for differential protection of power transformers," *Int. J. Electron. Electr. Power*, vol. 7, no. 2, pp. 51–61, 2019.

[27] Y. Yubo, C. Deshu, X. Zhenyu, and O. P. Malik, “Adaptive variable window algorithm for power transformer differential protection,” *IEEE Trans. Power Del.*, vol. 20, no. 3, pp. 1790–1798, 2005.

[28] S. Hayder, K. S. R. Rao, and S. S. Yadav, “Universal adaptive differential protection for regulating transformers,” *IEEE Trans. Power Del.*, vol. 23, no. 4, pp. 1976–1983, 2008.

[29] M. Bukhari, M. Asghar, and A. Ahmad, “Empirical wavelet transform based intelligent protection scheme for microgrids,” *Energies*, vol. 15, no. 14, p. 5095, 2022.

[30] S. Murugan, S. Simon, K. Venkatesh, and N. K. Sharma, “An efficient approach for power transformer differential protection using energy feature tracking,” *IEEE Trans. Power Del.*, vol. 32, no. 4, pp. 1850–1859, 2017.

[31] K. Hegazy and M. M. A. Salama, “Advanced differential protection scheme for power transformers based on wavelet packet transform,” *IEEE Trans. Power Del.*, vol. 35, no. 5, pp. 2345–2356, 2020.

[32] B. Kasztenny and E. Rosolowski, “Adaptive neural network-based protection of power transformers,” *Electr. Power Syst. Res.*, vol. 155, pp. 368–379, 2018.

[33] Z. Gajic, I. Brnivic, and R. Gojko, “Differential protection of power transformers using ICT and smart processing,” *Int. J. Electr. Power Energy Syst.*, vol. 110, pp. 582–592, 2019.

[34] D. R. Oliveira, A. S. Bretas, and L. U. Iurinic, “Transformers differential protection using independent component analysis,” *IEEE Trans. Power Del.*, vol. 34, no. 2, pp. 738–747, 2019.

[35] A. Hooshyar, M. A. Azzouz, and E. F. El-Saadany, “Transformer differential protection based on GARCH modelling of differential current,” *IEEE Trans. Power Del.*, vol. 31, no. 1, pp. 127–136, 2016.

[36] Y. C. Kang, B. E. Lee, and S. H. Kang, “A CT saturation detection algorithm using the secondary current third-derivative function,” *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2863–2871, 2011.

[37] X. Lin, P. Liu, and O. P. Malik, “Studies for identification of the inrush based on improved correlation algorithm,” *IEEE Trans. Power Del.*, vol. 17, no. 4, pp. 901–907, 2002.

APPENDIX A

SYSTEM PARAMETERS AND CONFIGURATION

The transformer and CT parameters correspond to the 300 MVA, 230/11 kV, Yd1 unit of Chapter 4 (Tables 4.1 and 4.2). Table A.1 lists the Daubechies-4 filter coefficients, and Table A.2 the LSTM hyperparameter search space.

Table A.1: Daubechies-4 decomposition filter coefficients ($g[n] = (-1)^n h[7-n]$).

n	$h[n]$ (low-pass)	$g[n]$ (high-pass)
0	$+0.23037781$	$+0.01059740$
1	$+0.71484657$	-0.03278364
2	$+0.63088077$	-0.03084138
3	-0.02798377	$+0.18703481$
4	-0.18703481	$+0.03084138$
5	$+0.03084138$	$+0.03278364$
6	$+0.03278364$	-0.01059740
7	-0.01059740	-0.02798377

Table A.2: LSTM hyperparameter search space and selected configuration.

Hyperparameter	**Search space**	**Selected**
LSTM layers	{1, 2, 3}	2
Hidden units (L1 / L2)	{64, 128, 256}	128 / 64
Dropout	{0.1, 0.2, 0.3, 0.5}	0.3
Learning rate	{0.01, 0.001, 0.0001}	0.001
Optimizer	{Adam, RMSprop, SGD}	Adam
Batch size	{32, 64, 128}	128
Early-stopping patience	{10, 15, 20, 25}	20
ONNX opset	{11, 13, 14}	14

APPENDIX B

REPRODUCIBILITY AND STANDARDS

The pipeline uses MATLAB R2023a/Simulink/Simscape Electrical (10 μs discrete solver), Python 3.11 with PyWavelets, PyTorch 2.x (Adam, weighted categorical cross-entropy), ONNX Opset 14, and the Simulink Predict block, with fixed random seeds. Relevant standards include IEEE C37.91 (transformer protection), IEEE C37.110 (CTs for relaying), IEC 60255 (measuring relays), IEC 61850-9-2 (Sampled Values; the 1.6 kHz interface emulated here), IEC 61850-8-1 (GOOSE), and IEC 62351 (communication security). Retaining a certified adaptive 87T element as the dependability anchor keeps the scheme aligned with IEEE C37.91, easing certification relative to a pure black-box classifier.

APPENDIX C

EXTENDED PER-SCENARIO RESULTS

This appendix disaggregates the test-set performance of Chapter 6. Table C.1 reports internal-fault recall by fault type (and the correct no-trip rate for the non-fault categories), confirming uniformly high detection across fault types. Table C.2 gives the end-to-end clearing-time breakdown that sums to the 17.2 ms average reported in Section 6.4.

Table C.1: Per-category recall of the proposed scheme (internal-fault types: trip recall; non-fault categories: correct no-trip rate).

Category	**Recall (%)**
AG (SLG)	99.7
BG (SLG)	99.5
CG (SLG)	99.7
ABG / ACG / BCG (LLG)	99.3
ABC (LLL)	99.8
ABCG (LLLG)	99.8
Magnetizing inrush (no-trip)	97.4
External fault with CT saturation (no-trip)	96.1

Table C.2: End-to-end clearing-time breakdown of the hybrid relay.

Stage	**Duration (ms)**
Data buffering (quarter-cycle)	5.0
DWT decomposition and energy computation	$<$1.0
LSTM inference (ONNX Predict block)	$<$1.0
Adaptive 87T computation	$<$1.0
Hybrid veto/AND-gate logic	$<$0.1
Breaker and pickup latency	$\approx$9.1
Total (average)	17.2

APPENDIX D

SUMMARY OF KEY DESIGN PARAMETERS

Table D.1: Consolidated summary of the principal design parameters of the proposed framework.

Parameter	**Value**
Protected transformer	300 MVA, 230/11 kV, Yd1
Relay sampling rate	1.6 kHz (32 samples/cycle)
EMT solver step	10 µs (discrete), MATLAB R2023a
DWT	3-level db4, 16-sample half-cycle window
Feature vector	6-D (E_A, E_D per phase)
LSTM	2 layers (128→64) + global temporal attention
Trainable parameters	125,476
Optimizer / loss	Adam ($\eta = 0.001$) / weighted categorical cross-entropy (internal-class weight 1.4155)
Batch size / max epochs	128 / 200 (early-stop patience 20)
Confidence threshold	$\theta_{conf} = 0.85$ (state-adaptive)
ONNX opset	14
Dataset	2,500 base cases (1,035/578/500/387); 70/15/15 split; 1,800-case test
Headline results	98.89% classifier acc (F1 0.9901, AUC 0.9999); 100% hybrid dep. & sec.; 17.2 ms clearing

APPENDIX E

CODE LISTINGS

This appendix provides representative code excerpts of the implementation pipeline: DWT energy-feature extraction in MATLAB, LSTM training and ONNX export in Python/PyTorch, and the Simulink deployment configuration.

E.1 MATLAB: DWT Energy-Feature Extraction

```
function F = dwt_features(I_diff, fs)
% I_diff : 3xN differential currents (phases A,B,C); fs = 1600 Hz
% Returns : [EA_a ED_a EA_b ED_b EA_c ED_c]  (6 energy features)
    w = 'db4';
    n = floor(fs/(2*50));            % 16-sample half-cycle window
    F = zeros(1,6);
    for ph = 1:3
        x = I_diff(ph, end-n+1:end);
        [C, L] = wavedec(x, 3, w);   % 3-level db4 decomposition
        a3 = appcoef(C, L, w, 3);    % approximation (0-100 Hz)
        d  = detcoef(C, L, [1 2 3]); % detail bands (100-800 Hz)
        EA = sum(a3.^2);
        ED = sum(d{1}.^2) + sum(d{2}.^2) + sum(d{3}.^2);
        F(2*ph-1) = EA;  F(2*ph) = ED;
    end
end
```

E.2 Python/PyTorch: LSTM Training and ONNX Export (Opset 14)

```
import numpy as np, torch, torch.nn as nn
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X = np.load('dwt_features.npy')      # (N, 32, 6)
y = np.load('labels.npy')            # 4-class: 0 normal, 1 external, 2 inrush, 3 internal
```

```python
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.15, stratify=y)
sc = StandardScaler()
Xtr = sc.fit_transform(Xtr.reshape(-1,6)).reshape(-1,32,6)
Xte = sc.transform(Xte.reshape(-1,6)).reshape(-1,32,6)

class DWTLSTM(nn.Module):
    def __init__(self):
        super().__init__()
        self.l1 = nn.LSTM(6, 128, batch_first=True)     # layer 1
        self.d1 = nn.Dropout(0.3)
        self.l2 = nn.LSTM(128, 64, batch_first=True)    # layer 2
        self.d2 = nn.Dropout(0.3)
        self.attn = nn.Linear(64, 1)                    # temporal attention
        self.fc1, self.fc2 = nn.Linear(64, 32), nn.Linear(32, 4)
    def forward(self, x):
        h,_ = self.l1(x); h = self.d1(h)
        h,_ = self.l2(h); h = self.d2(h)
        a = torch.softmax(self.attn(h), dim=1)          # (B,32,1)
        c = (a * h).sum(dim=1)                          # context (B,64)
        return self.fc2(torch.relu(self.fc1(c)))

model = DWTLSTM()
opt = torch.optim.Adam(model.parameters(), lr=1e-3)
# class weights: internal-fault (Trip) class emphasised (pos. weight 1.4155)
w = torch.tensor([1.0, 1.0, 1.0, 1.4155])
loss_fn = nn.CrossEntropyLoss(weight=w)
# protection decision: Trip iff argmax(probs) == 3 (internal fault)
# ... training loop: 200 epochs max, batch 128, early stopping patience 20 ...

dummy = torch.randn(1, 32, 6)
torch.onnx.export(model, dummy, 'dwt_lstm_relay.onnx',
                  input_names=['input'], output_names=['probs'],
                  opset_version=14)
```

E.3 Simulink: ONNX Predict and Hybrid-Gate Configuration

```matlab
%% EMT solver at 10 us; relay subsystem decimated to 1.6 kHz
set_param('hatdp_relay','SolverType','Fixed-step', ...
          'Solver','FixedStepDiscrete','FixedStep','1e-5','StopTime','1.0');

%% ONNX Predict block (Opset 14 model)
set_param('hatdp_relay/LSTM_Classifier', ...
```

```
        'NetworkFilename','dwt_lstm_relay.onnx', ...
        'OutputLayer','softmax','ExecutionMode','SIM');

%% Adaptive 87T relay and hybrid AND/veto gate
relay.Slope = 0.30;  relay.Ipick = 0.25;          % pu of rated
relay.I2_restraint = 0.20;  relay.CrossBlock = 'on';
gate.Mode = 'AND';  gate.LSTM_class = 3;          % class 3 = internal fault (Trip)
gate.ConfThreshold = 0.85;
```